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SHEET IDENTIFICATION DEVICE, IMAGE PROCESSING APPARATUS, AND SHEET IDENTIFICATION METHOD

Abstract

A sheet identification device includes an acquisition portion and an asperity identification portion. The acquisition portion acquires an identification image (Im1). The identification image (Im1) is an image of an identification region (R1) of the surface (A1) of a sheet (Sh1) on which image formation or image reading is performed. The identification region (R1) is a region on which pattern light (P1) is projected of the surface (A1) of the sheet (Sh1). The asperity identification portion identifies asperity information on the asperities on the surface (A1) of the sheet (Sh1), based on the identification image (Im1).

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a sheet identification device, an image processing apparatus, and a sheet identification method.

BACKGROUND

[0002] As a related technique, there is known a technique used in an image forming apparatus such as a copier or a laser printer, which automatically identifies the type of a sheet (paper) from an image of the surface of the sheet (see, for example, Patent Literature 1). An image reading apparatus according to the related technique is provided with a light emitting element that obliquely irradiates the surface of the sheet with light, and an area sensor that reads the irradiation region as an image, and reads information on the sheet from the reading result.

[0003] In this image reading apparatus, the surface roughness of the sheet is estimated by detecting a shadow image caused by the asperities on the surface of the sheet from the image of the irradiation region. When the asperities on the surface of the sheet are large, the contrast is higher than when the asperities are small, so that the magnitude of the asperities on the surface can be estimated from the contrast. Further, this image reading apparatus is configured to set the incident direction of the light from the light emitting element at an angle of 45 degrees with respect to the conveying direction of the sheet so as to maintain the fiber direction of the sheet and the incident direction of the light at an angle of approximately 45 degrees and reduce the variation in the detection accuracy depending on the fiber direction.

CITATION LIST

Patent Literature

[0004] Patent Literature 1: Japanese Unexamined Patent Application Publication No. 2004-038879

SUMMARY OF INVENTION

Technical Problem

[0005] However, with the above-described configuration of the related technique, it is necessary to make the angle of the light incident direction with respect to the surface of the sheet shallow (small) in order to obtain an image having a high sensitivity to asperities, and thus the obtained image becomes dark as a whole, and the shadow caused by the asperities is easily buried in noise.

[0006] An object of the present invention is to provide a sheet identification device, an image processing apparatus, and a sheet identification method which can easily improve the accuracy of identifying the asperities of the surface of the sheet.

Solution to Problem

[0007] A sheet identification device according to one aspect of the present invention includes an acquisition portion and an asperity identification portion. The acquisition portion acquires an identification image which is an image of an identification region. The identification region is a region on which pattern light is projected of the surface of a sheet on which image formation or image reading is performed. The asperity identification portion identifies asperity information on asperities on the surface of the sheet, based on the identification image.

Advantageous Effects of Invention

[0008] According to the present invention, it is possible to provide a sheet identification device, an

image processing apparatus, and a sheet identification method which can easily improve the accuracy of identifying the asperities on the surface of the sheet.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- [0009] FIG. 1 is a schematic block diagram of an image processing apparatus according to a first embodiment.
- [0010] FIG. 2 is a schematic diagram showing the exterior and internal configuration of the image processing apparatus according to the first embodiment.
- [0011] FIG. 3 is a schematic diagram showing a light irradiation portion and an imaging portion of a sheet identification device according to the first embodiment.
- [0012] FIG. 4 is a schematic diagram showing a sheet and the light irradiation portion of the sheet identification device according to the first embodiment.
- [0013] FIG. 5 is a schematic diagram showing the principle of sheet surface asperity detection in the sheet identification device according to the first embodiment.
- [0014] FIG. 6 is a diagram showing an example of an identification image obtained in the sheet identification device according to the first embodiment.
- [0015] FIG. 7 is a flowchart of an example operation of the sheet identification device according to the first embodiment.
- [0016] FIG. 8 is a diagram showing an example of the identification image when a predetermined angle obtained in the sheet identification device according to the first embodiment is changed.
- [0017] FIG. 9 is a graph showing the relationship between the arithmetic average height and the standard deviation obtained in the sheet identification device according to the first embodiment.
- [0018] FIG. 10 is a table showing the results of calculating the determination coefficient while changing the line width of pattern light and the relationship between the irradiation direction of the pattern light and the fiber direction in the sheet identification device according to the first embodiment.
- [0019] FIG. 11 is a schematic diagram showing pattern light which produces a lattice pattern in the sheet identification device according to the first embodiment.
- [0020] FIG. 12 is a schematic block diagram of an image processing apparatus according to a second embodiment.
- [0021] FIG. 13 is a schematic diagram showing a sheet and a light irradiation portion of a sheet identification device according to a third embodiment.
- [0022] FIG. 14 is a diagram showing an example of an identification image and an integrated image obtained in the sheet identification device according to the third embodiment.
- [0023] FIG. 15 is a diagram showing an example of the identification image and the integrated image obtained in the sheet identification device according to the third embodiment.
- [0024] FIG. 16 is a graph showing pixel values of pixels, focusing on one row of the integrated image obtained in the sheet identification device according to the third embodiment.
- [0025] FIG. 17 is a graph showing the relationship between the arithmetic average height and the width of an intermediate layer obtained in the sheet identification device according to the third embodiment.
- [0026] FIG. 18 is a flowchart of an example operation of the sheet identification device according to the third embodiment.

DETAILED DESCRIPTION

[0027] Embodiments of the present invention will be described below with reference to the accompanying drawings. The following embodiments are examples of embodying the present invention and do not limit the technical scope of the present invention.

First Embodiment

[1] Overall Configuration of Image Processing Apparatus

[0028] First, an overall configuration of an image processing apparatus **10** according to the present embodiment will be described with reference to FIG. **1** and FIG. **2**.

[0029] The image processing apparatus **10** according to the present embodiment is a multifunction peripheral having a plurality of functions such as a scanning function for reading an image (image data) from a document sheet, a printing function for forming an image based on the image data, a facsimile function, and a copy function. The image processing apparatus **10** may be a printer, a scanner, a facsimile machine, a copier, or the like as long as it has an image processing function including at least one of a function of forming an image and a function of reading an image.

[0030] As shown in FIG. **1**, the image processing apparatus **10** includes an automatic document conveying device **11**, an image reading portion **12**, an image forming portion **13**, a sheet feed portion **14**, an operation display portion **15**, and a control portion **16**. Since the automatic document conveying device **11** is an auto document feeder (ADF), it is indicated as “ADF” in FIG. **1** and referred to as an “ADF **11**” in the following description. In the present embodiment, as shown in FIG. **2**, the image processing apparatus **10** includes a housing **100**. The ADF **11**, the image reading portion **12**, the image forming portion **13**, the sheet feed portion **14**, the operation display portion **15**, and the control portion **16** are provided in the housing **100**.

[0031] The ADF **11** conveys a sheet (document sheet) whose image is read by the image reading portion **12**. The ADF **11** includes a document sheet loading portion, a plurality of conveying rollers, a document sheet holder, a sheet discharge portion, and the like.

[0032] The image reading portion **12** reads an image from a sheet and outputs image data corresponding to the read image. The image reading portion **12** includes a document sheet table, a light source, a plurality of mirrors, an optical lens, a charge coupled device (CCD), and the like.

[0033] The image forming portion **13** forms an image on a sheet Sh1 based on the image data output from the image reading portion **12** (see FIG. **2**). In addition, the image forming portion **13** forms an image on a sheet Sh1 based on image data input from an information processing apparatus, such as a personal computer, external to the image processing apparatus **10**. In the present embodiment, as an example, as shown in FIG. **2**, the image forming portion **13** includes a transfer device **131**, a fixing device **132**, a sheet discharge tray **133**, and the like, and forms an image on a sheet Sh1 using an electrophotographic method. The image forming portion **13** is not necessarily configured to form only monochrome images, and may be configured to form full-color images using four colors of C (cyan), M (magenta), Y (yellow), and K (black). In addition, the image forming portion **13** may form an image on a sheet using an image forming method other than the electrophotographic method, such as an inkjet method.

[0034] The image forming portion **13** forms an image on a sheet Sh1 using toner as a developer. Specifically, the image forming portion **13** forms an electrostatic latent image on a charged surface of the photoconductor drum by irradiating the surface with a laser beam, and forms a toner image on the surface of the photoconductor drum by developing the electrostatic latent image with toner. The transfer device **131** transfers the toner image to the sheet Sh1 conveyed through a conveying path T1 (see FIG. **2**). The fixing device **132** melts and fixes the toner image transferred to the sheet Sh1 onto the sheet Sh1. For example, the fixing device **132** includes a fixing roller and a pressure roller, and heats the toner image transferred to the sheet Sh1 and presses the sheet Sh1, thereby fixing the toner image onto the sheet Sh1. The sheet Sh1 on which the image has been formed is discharged to the sheet discharge tray **133**. When the image forming portion **13** forms an image using an inkjet method, ink (another example of the developer) is supplied instead of the toner.

[0035] The sheet feed portion **14** supplies the sheet Sh1 to the image forming portion **13**. The sheet feed portion **14** includes a plurality of sheet feed cassettes **141**, a manual feed tray, and a plurality of conveying rollers. The sheet feed portion **14** conveys the sheet Sh1 from the plurality of sheet feed cassettes **141**, the manual feed tray, or the like through the conveying path T1 by the plurality

of conveying rollers and the like to supply the sheet Sh1 to the image forming portion **13**. The image forming portion **13** forms an image on the sheet Sh1 supplied from the sheet feed portion **14** through the conveying path T1.

[0036] The operation display portion **15** is a user interface in the image processing apparatus **10**. The operation display portion **15** includes a display portion, such as a liquid crystal display, for displaying various types of information in response to a control instruction from the control portion **16**, and an operation portion, such as a switch or a touch panel, for inputting various types of information to the control portion **16** in response to a user's operation. In addition, the image processing apparatus **10** may include, as a user interface, an audio output portion, an audio input portion, and the like, in addition to or instead of the operation display portion **15**.

[0037] The control portion **16** comprehensively controls the image processing apparatus **10**. The control portion **16** is mainly composed of a computer system including one or more processors and one or more memories. In the image processing apparatus **10**, the functions of the control portion **16** are realized by one or more processors executing programs. The programs may be stored in advance in the one or more memories, may be provided through a telecommunications line such as the Internet, or may be provided by being stored in on a non-transitory recording medium readable by the computer system, such as a memory card or an optical disk. The one or more processors are composed of one or more electronic circuits, including a semiconductor integrated circuit. Further, the computer system in the present disclosure includes a microcontroller having one or more processors and one or more memories. The control portion **16** may be a control portion provided separately from the main control portion which comprehensively controls the image processing apparatus **10**.

[0038] In addition, the image processing apparatus **10** further includes a storage portion, a communication portion, a power supply portion, and the like. The storage portion includes one or more nonvolatile memories, and stores in advance information, such as control programs, for causing the control portion **16** to execute various types of processing. The communication portion is an interface that executes data communication between the image processing apparatus **10** and an external apparatus connected via a communication network such as the Internet or a local area network (LAN). The power supply portion is a power supply circuit that generates (outputs) electric power for the operation of the image processing apparatus **10**.

[0039] As a technique related to this type of image processing apparatus **10**, there is known a technique which is used in an image forming apparatus such as a copier or a laser printer to automatically identify the type of a sheet (paper) from an image of the surface of the sheet. An image reading apparatus according to the related technique includes a light emitting element that obliquely irradiates the surface of the sheet with light, and an area sensor that reads the irradiation region as an image, and reads information on the sheet from the read result.

[0040] In this image reading apparatus, the surface roughness of the sheet is estimated by detecting a shadow image caused by the asperities on the surface of the sheet from the image of the irradiation region. When the asperities on the surface of the sheet are large, the contrast is higher than when the asperities are small, so that the magnitude of the asperities on the surface can be estimated from the contrast. Further, this image reading apparatus is configured to set the incident direction of the light from the light emitting element at an angle of 45 degrees with respect to the conveying direction of the sheet so as to maintain the fiber direction of the sheet and the incident direction of the light at an angle of approximately 45 degrees and reduce the variation in the detection accuracy depending on the fiber direction.

[0041] However, with the above-described configuration of the related technique, it is necessary to make the angle of the light incident direction with respect to the surface of the sheet shallow (small) in order to obtain an image having a high sensitivity to asperities, and thus the obtained image becomes dark as a whole, and the shadow caused by the asperities is easily buried in noise.

[0042] In contrast, in the present embodiment, the image processing apparatus **10** can easily

improve the accuracy of identifying the asperities on the surface of the sheet with the configuration to be described below.

[0043] That is, as shown in FIG. 1, the image processing apparatus **10** according to the present embodiment includes a sheet identification device **2**. The sheet identification device **2** according to the present embodiment is integrated with the image processing apparatus **10**.

[0044] The sheet identification device **2** includes an acquisition portion **21** and an asperity identification portion **22**. The acquisition portion **21** acquires an identification image Im**1** (see FIG. 4). The identification image Im**1** is an image of an identification region R**1** (see FIG. 4) of the surface A**1** (see FIG. 3) of the sheet Sh**1** on which image formation or image reading is performed. The identification region R**1** is a region on the surface A**1** of the sheet Sh**1** on which pattern light P**1** (see FIG. 3) is projected. The asperity identification portion **22** identifies asperity information on the asperities on the surface A**1** of the sheet Sh**1** based on the identification image Im**1**. In the present embodiment, the acquisition portion **21** and the asperity identification portion **22**, which are constituent elements of the sheet identification device **2**, are provided in the control portion **16** as functions of the control portion **16**.

[0045] With the above configuration, the sheet identification device **2** according to the present embodiment and the image processing apparatus **10** provided with the sheet identification device **2** have an advantage that the accuracy of identifying the asperities on the surface A**1** of the sheet Sh**1** can be easily improved. In other words, the identification region R**1** of the surface A**1** of the sheet Sh**1** is not uniformly irradiated with the light from the light emitting element, but rather the pattern light P**1** is projected thereon. Therefore, the asperity identification portion **22** can identify the asperity information on the asperities on the surface A**1** of the sheet Sh**1** from the degree of deformation or distortion of the pattern light P**1** in the identification image Im**1**. Therefore, the asperity information can be identified from a relatively bright identification image Im**1** without making the angle of the light incident direction with respect to the surface A**1** of the sheet Sh**1** shallow (small) as in the related technique, and as a result, the accuracy of identifying the asperities can be easily improved as compared with the related technique.

[0046] The sheet identification device **2** according to the present embodiment constitutes the image processing apparatus **10** together with the image processing portion (the image reading portion **12** and the image forming portion **13**). In other words, the image processing apparatus **10** according to the present embodiment includes the sheet identification device **2** and an image processing portion that executes at least one of image formation and image reading on the sheet Sh**1**.

[2] Definitions

[0047] The “sheet” in the present disclosure is a sheet on which image formation or image reading is to be performed. In the present embodiment, as an example, it is assumed that the sheet Sh**1** to be irradiated with the pattern light P**1** is a sheet Sh**1** on which image formation is to be performed by the image forming portion **13**. That is, in the present embodiment, the sheet Sh**1** conveyed through the conveying path T**1** by the sheet feed portion **14** is to be irradiated with the pattern light P**1**. However, the present disclosure is not limited to this example, and the sheet to be irradiated with the pattern light P**1** may be the sheet (document sheet) on which image reading is to be performed by the image reading portion **12**, that is, the sheet conveyed by the ADF **11**. In addition, although the sheet Sh**1** is paper as an example in the present embodiment, it is not limited to paper, and may be, for example, a resin film.

[0048] The “pattern light” in the present disclosure is, for example, light that is projected from a light irradiation portion **3** (see FIG. 1) onto a projection surface (here, the surface A**1** of the sheet Sh**1**) in a controlled shape and direction, and is so-called structured light. That is, the region (identification region R**1**) irradiated with the pattern light P**1** is not uniformly irradiated with the pattern light P**1**, but a figure, a design, a picture, a pattern, a symbol, a character, a number, or the like corresponding to the pattern light P**1** is projected thereon. Specifically, by irradiating the identification region R**1** with the pattern light P**1**, a luminance distribution of a pattern

corresponding to the pattern light P1, such as a stripe pattern, a lattice pattern, or an arc pattern, is produced in the identification region R1. Further, the pattern light P1 is not limited to a fixed pattern such as a still image, and a luminance distribution of a pattern that changes with time such as a moving image (including an animation) may be produced in the identification region R1.

[0049] The “identification image” in the present disclosure is, for example, an image of the identification region R1 on which the pattern light P1 is projected, which is captured by an imaging portion 4. That is, the identification image Im1 includes the pattern light P1 projected onto the identification region R1, or more strictly, a luminance distribution of a pattern corresponding to the pattern light P1 produced in the identification region R1 by projecting the pattern light P1 on the identification region R1. The identification image Im1 may be either a monochrome image or a color image, and may be either a still image or a moving image.

[0050] The “asperity information” in the present disclosure is information on the asperities on the surface A1 of the sheet Sh1, and includes information such as the height (or depth) of the asperities and/or the size of the asperities in plan view. The surface (A1) of the sheet (Sh1) has asperities including at least one of a concave portion and a convex portion. That is, the surface A1 may include only a plurality of concave portions or a plurality of convex portions. Further, the surface A1 may include a plurality of concave portions and one convex portion. In this case, as an example, the surface A1 includes one net-like convex portion and a plurality of concave portions consisting of mesh portions surrounded by this convex portion. Similarly, as an example, the surface A1 may include one net-like concave portion and a plurality of convex portions consisting of mesh portions surrounded by this concave portion.

[0051] The asperities (concave portions and convex portions) of the surface A1 have extremely small sizes that cannot be individually identified with the naked eye, and the surface A1 of one sheet Sh1 includes a large number of asperities. That is, the asperities are microscopic compared to the entire surface A1, and when a person looks at the surface A1, the asperities make the surface A1 look like a rough “satin finish”. Such a large number of microscopic asperities are formed, for example, by a large number of fibers constituting paper when the sheet Sh1 is paper, or by embossing or the like when the sheet Sh1 is a resin film. Information on such microscopic asperities includes an index representing surface roughness, such as an arithmetic average height (Sa) or an arithmetic average height of lines (Ra).

[0052] The “fiber direction” in the present disclosure is the direction of the fibers on the surface A1 of the sheet Sh1, and is, for example, the extending direction of a large number of fibers constituting paper when the sheet Sh1 is paper, i.e., the paper grain direction. Generally, the sheet Sh1 has a “long grain” in which the fiber direction is along the long side of the sheet Sh1, and a “short grain” in which the fiber direction is along the short side of the sheet Sh1. The conveying direction D1 (see FIG. 2) of the sheet Sh1 in the image processing apparatus 10 is a direction along the long side or the short side of the sheet Sh1. Therefore, basically, the fiber direction is along the conveying direction D1 of the sheet Sh1 or along a direction orthogonal to the conveying direction D1.

[3] Sheet Identification Device

[0053] Next, a configuration of the sheet identification device 2 according to the present embodiment will be described in more detail with reference to FIG. 1 to FIG. 4.

[0054] In the present embodiment, the sheet identification device 2 includes an acquisition portion 21, an asperity identification portion 22, a condition determination portion 23, a direction identification portion 24, a thickness identification portion 25, a light irradiation portion 3, an imaging portion 4, and a thickness sensor 5. The acquisition portion 21, the asperity identification portion 22, the condition determination portion 23, the direction identification portion 24, and the thickness identification portion 25 are provided in the control portion 16 as functions of the control portion 16. That is, in the present embodiment, the image processing apparatus 10 includes not only the acquisition portion 21, the asperity identification portion 22, but also the condition

determination portion **23**, the direction identification portion **24**, and the thickness identification portion **25**, as functions of the control portion **16**.

[0055] The light irradiation portion **3** irradiates the surface **A1** of the sheet **Sh1** with the pattern light **P1**. That is, the light irradiation portion **3** generates pattern light **P1** whose shape and direction are controlled, and irradiates the surface **A1** of the sheet **Sh1** with the pattern light **P1**, thereby projecting the pattern light **P1** on the identification region **R1** of the surface **A1** of the sheet **Sh1**. With such pattern light **P1** from the light irradiation portion **3**, a figure, a design, a picture, a pattern, a symbol, a character, a number, or the like corresponding to the pattern light **P1** is projected on the identification region **R1** of the surface **A1** of the sheet **Sh1**.

[0056] In the present embodiment, as an example, the pattern light **P1** forms a stripe pattern in which a bright portion **L1** and a dark portion **L2** are alternately arranged on the identification region **R1**, as shown in FIG. **4**. That is, the projection of the pattern light **P1** produces a luminance distribution of a stripe pattern including the bright portions **L1** and the dark portions **L2** on the identification region **R1**. Here, the bright portion **L1** is a region brighter than the dark portion **L2**, and in other words, the stripe pattern is a pattern in which a plurality of bright portions **L1** are arranged at intervals or a pattern in which a plurality of dark portions **L2** are arranged at intervals. That is, in the identification region **R1**, the pattern light **P1** is projected to form a stripe pattern in which a linear bright portion **L1** and a linear dark portion **L2** are alternately arranged in a direction orthogonal to the longitudinal directions of the linear bright portion **L1** and the linear dark portion **L2**. In the present embodiment, as an example, a linear bright portion **L1** and a linear dark portion **L2**, which are orthogonal to the conveying direction **D1** of the sheet **Sh1**, are arranged alternately in the conveying direction **D1**. In FIG. **4** and the like, the bright portion **L1** is indicated by shading (dot hatching), and the dark portion **L2** is indicated by blackening. Thus, deformation, distortion, or the like according to the asperities on the surface **A1** is likely to appear in the stripe pattern on the identification region **R1**. However, the pattern light **P1** that produces the stripe pattern as shown in FIG. **4** is merely an example of the pattern light **P1**, and the pattern light **P1** can be changed as appropriate.

[0057] In the present embodiment, as shown in FIG. **3** and FIG. **4**, the light irradiation portion **3** includes a light source **31** and a shield **32**. The shield **32** blocks part of the light output from the light source **31** to allow the pattern light **P1** to pass therethrough. The light source **31** includes a light emitting element that emits light when electric power is supplied thereto, and outputs light generated by the light emitting element toward the shield **32**. The light source **31** is controlled by a control signal from the control portion **16**, and can be switched on/off at least at the control portion **16**. The shield **32** is disposed between the light source **31** and the identification region **R1** of the surface **A1** of the sheet **Sh1**, and is a component that blocks part of the light from the light source **31** and allows the rest to pass therethrough. Thus, part of the light output from the light source **31** is blocked by the shield **32**, and the rest passes through the shield **32**, so that the light that has passed through the shield **32** becomes pattern light **P1** controlled to have a desired shape by the shield **32**. Thus, the pattern light **P1** can be realized with a relatively simple configuration.

[0058] In the present embodiment, as an example, the light source **31** has one or more light emitting elements such as a light emitting diode (LED) or an organic electroluminescence (EL), and makes the entire light emitting surface **311** (see FIG. **3**) in a rectangular shape in plain view to substantially uniformly emit light to perform surface emission. Further, in the present embodiment, the light source **31** outputs parallel light or light close to parallel light. Therefore, the optical axis **Ax1** of the pattern light **P1** output from the light irradiation portion **3** (see FIG. **3**) is a perpendicular line of the light emitting surface **311** passing through the center (center of gravity) of the light emitting surface **311** of the light source **31**. The light source **31** may have an optical component, such as a collimator lens, for converting the light from the light emitting element into parallel light. In the present embodiment, as an example, the light source **31** outputs visible light, specifically, white light. However, the light source **31** only has to output light having a wavelength

to which the imaging portion 4 is sensitive, and may output light other than white light, or may output light in a wavelength range other than visible light, such as infrared light or ultraviolet light. [0059] In the present embodiment, as an example, the shield 32 is a rectangular plate-shaped component that absorbs or reflects light from the light source 31, and has one or more slits 321 (see FIG. 4). Thus, part of the light output from the light source 31 is blocked by the shield 32, and the rest passes through the shield 32 through the slit 321 of the shield 32. In the present embodiment, in order to realize the pattern light P1 that forms a stripe pattern in the identification region R1, the shield 32 has a plurality of linear slits 321. Thus, in the identification region R1 irradiated with the pattern light P1, the light that has passed through the slit 321 becomes the bright portion L1, and the shadow of the shield 32 becomes the dark portion L2, so that the stripe pattern is projected. However, the present disclosure is not limited to the configuration in which the light that has passed through the slit 321 becomes the bright portion L1 as described above, and the pattern light P1 may be realized by, for example, “interference stripes” utilizing interference of light.

[0060] Here, a first imaginary straight line connecting the light irradiation portion 3 that applies the pattern light P1 and the center of the identification region R1 is inclined at a predetermined angle $\theta 1$ with respect to a second imaginary straight line extending along the conveying direction D1 of the sheet Sh1. In the present embodiment, in the identification region R1, the surface A1 of the sheet Sh1 is along the conveying direction D1 of the sheet Sh1, so that the angle between the first imaginary straight line and the surface A1 of the sheet Sh1 is the predetermined angle $\theta 1$. Further, since the first imaginary straight line is the optical axis Ax1 of the pattern light P1, the optical axis Ax1 of the pattern light P1 is inclined at the predetermined angle $\theta 1$ with respect to the surface A1 of the sheet Sh1. In particular, in the present embodiment, the light irradiation portion 3 is configured to irradiate the identification region R1 with the pattern light P1 obliquely at the predetermined angle $\theta 1$ from the downstream side in the conveying direction D1, that is, the front side in the conveying direction of the sheet Sh1. Thus, deformation, distortion, or the like according to the asperities on the surface A1 is likely to appear in the pattern on the identification region R1.

[0061] The imaging portion 4 captures an image of the identification region R1 of the surface A1 of the sheet Sh1 as the identification image Im1. Since the image captured by the imaging portion 4 is an image (identification image Im1) of the identification region R1 on which the pattern light P1 is being projected, the light irradiation portion 3 irradiates the identification region R1 with the pattern light P1 at least at the imaging timing of the imaging portion 4. In the present embodiment, as an example, the imaging portion 4 and the light irradiation portion 3 are synchronized with each other, and the light irradiation portion 3 applies the pattern light P1 in accordance with the imaging timing of the imaging portion 4. In other words, the light irradiation portion 3 does not output the pattern light P1 during the period in which the imaging portion 4 does not perform imaging, thereby suppressing unnecessary power consumption in the light irradiation portion 3.

[0062] In the present embodiment, as shown in FIG. 3, the imaging portion 4 includes an imaging element 41 and an optical component 42. The imaging element 41 includes an area sensor or a line sensor, and outputs image data of the captured identification image Im1 to the control portion 16 as an electric signal. In the present embodiment, as an example, the imaging portion 4 is an area sensor of a contact image sensor (CIS) system using a complementary metal oxide semiconductor (CMOS) sensor as the imaging element 41. However, the imaging portion 4 is not limited to this example, and may be a sensor of a CCD system using a charge coupled device (CCD) as the imaging element 41, for example.

[0063] The optical component 42 includes, for example, an imaging lens, and is disposed between the imaging element 41 and the identification region R1 of the surface A1 of the sheet Sh1. Thus, the light of the identification region R1 enters the imaging element 41 through the optical component 42. In the present embodiment, the imaging element 41 and the optical component 42 are arranged on a perpendicular line of the identification region R1 passing through the center

(center of gravity) of the identification region R1. Further, a light receiving surface 411 (see FIG. 3) of the imaging element 41 is arranged parallel to the identification region R1. Therefore, the optical axis Ax2 (see FIG. 3) of the imaging portion 4 is a perpendicular line of the light receiving surface 411 passing through the center (center of gravity) of the light receiving surface 411 of the imaging element 41. The optical axis Ax2 of the imaging portion 4 is also orthogonal to the identification region R1, and intersects the optical axis Ax1 of the light irradiation portion 3 (pattern light P1) at the center of the identification region R1. The area of the identification region R1 captured by the imaging element 41 is equal to a numerical value obtained by dividing the area of the light receiving surface 411 of the imaging element 41 by the image magnification M of the optical component 42. In the present embodiment, the image magnification M is assumed to be "1" for simplicity of explanation. However, the image magnification M may be a value other than 1.

[0064] In the present embodiment, as an example, the imaging portion 4 is integrated with the light irradiation portion 3 to form a sensor unit 20 (see FIG. 2). In other words, the sensor unit 20 includes the light irradiation portion 3 and the imaging portion 4. The sensor unit 20 is housed in the housing 100 of the image processing apparatus 10, and is electrically connected to at least the control portion 16.

[0065] In the present embodiment, as shown in FIG. 2, the sensor unit 20 including the light irradiation portion 3 and the imaging portion 4 is arranged to face the conveying path T1 between the sheet feed portion 14 and the image forming portion 13. Therefore, the imaging position of the identification region R1 is set on the conveying path T1 between the sheet feed portion 14 and the image forming portion 13. That is, the light irradiation portion 3 and the imaging portion 4 can capture the identification image Im1 by irradiating the sheet Sh1 conveyed from the sheet feed portion 14 to the image forming portion 13 with the pattern light P1 at a position between the sheet feed portion 14 and the image forming portion 13. More specifically, the sensor unit 20 is disposed at a position upstream of the transfer device 131 of the image forming portion 13 and downstream of the merging point of the conveying path T1 connected to the plurality of sheet feed cassettes 141 in the conveying direction D1 of the sheet Sh1. Therefore, the identification image Im1 can be captured by one sensor unit 20 also for the sheets Sh1 supplied to the image forming portion 13 from the plurality of sheet feed cassettes 141, eliminating the need to provide a sensor unit 20 for each sheet feed cassette 141.

[0066] The surface A1 of the sheet Sh1 including the identification region R1 is a side where an image is formed by the image forming portion 13 in the thickness direction of the sheet Sh1, as an example in the present embodiment, but is not limited to this example. The identification region R1 may be set, for example, on a side (back side) where an image is not formed by the image forming portion 13 in the thickness direction of the sheet Sh1. In this case, the light irradiation portion 3 and the imaging portion 4 are disposed on the back side of the sheet Sh1. In addition, the identification region R1 may be set, for example, on both sides in the thickness direction of the sheet Sh1. In this case, two sets of the light irradiation portion 3 and the imaging portion 4 may be provided on both sides in the thickness direction of the sheet Sh1, or the sheet Sh1 may be turned over so that the identification images Im1 on both sides of the sheet Sh1 are captured by one set of the light irradiation portion 3 and the imaging portion 4.

[0067] The thickness sensor 5 detects a physical quantity relating to the thickness of the sheet Sh1. The thickness sensor 5 outputs the detected physical quantity as an electric signal to the control portion 16. Thus, the control portion 16 can identify the thickness of the sheet Sh1. As an example, the thickness sensor 5 includes an optical sensor that detects the thickness (or basis weight) of the sheet Sh1 using transmitted light. The thickness sensor 5 may be included in the sensor unit 20 or provided separately from the sensor unit 20.

[0068] The acquisition portion 21 acquires the identification image Im1 captured by the imaging portion 4. Specifically, the acquisition portion 21 acquires the image data of the identification image Im1 captured by the imaging portion 4 as an electric signal from the imaging element 41 of

the imaging portion **4**. The acquisition portion **21** controls the light irradiation portion **3** and the imaging portion **4** to cause the light irradiation portion **3** to apply the pattern light **P1** and the imaging portion **4** to capture the identification image **Im1** in accordance with, for example, the timing when the sheet **Sh1** passes a position on the conveying path **T1** corresponding to the sensor unit **20**. The identification image **Im1** acquired by the acquisition portion **21** is temporarily stored in the one or more memories. The acquisition portion **21** may acquire the identification image **Im1** from a source other than the imaging portion **4**.

[0069] The asperity identification portion **22** identifies asperity information on the asperities on the surface **A1** of the sheet **Sh1** based on the identification image **Im1** acquired by the acquisition portion **21**. Thus, the state of the asperities on the surface **A1** of the sheet **Sh1** can be identified. The asperity information includes information on at least one of the dimension of the asperities on the surface **A1** in a direction orthogonal to the plane along the surface **A1** and the dimension in a direction along the plane. That is, the asperity information includes information on the height (or depth) of the asperities, which is the dimension in the direction orthogonal to the plane along the surface **A1** and/or the size of the asperities in plan view, which is the dimension in the direction along the plane. Thus, the height (or depth) of the asperities on the surface **A1** of the sheet **Sh1** and/or the size of the asperities in plan view can be identified. In the present embodiment, as an example, the asperity identification portion **22** calculates a numerical value corresponding to the arithmetic average height (**Sa**) of the surface **A1** relating to the heights (or depths) of the asperities as the asperity information.

[0070] Here, the asperity identification portion **22** identifies the asperity information based on the degree of deformation, distortion, or the like of the pattern light **P1** in the identification image **Im1**. That is, since the identification image **Im1** includes a luminance distribution of a pattern (stripe pattern in the present embodiment) corresponding to the pattern light **P1**, which is produced in the identification region **R1** by the projection of the pattern light **P1**, the pattern is deformed or distorted by the asperities on the surface **A1**. For example, even when the pattern light **P1** forms a linear pattern, the pattern light **P1** projected on the surface **A1** is deformed (meandered) in accordance with the asperities on the surface **A1**. Therefore, the asperity identification portion **22** calculates asperity information on the asperities on the surface **A1** from the degree of deformation, distortion, or the like of the pattern light **P1**. In the present embodiment, the asperity identification portion **22** identifies the asperity information based at least on the variation in the line width of the pattern light **P1** on the identification region **R1**. Thus, the state of the asperities on the surface **A1** of the sheet **Sh1** can be identified by relatively simple arithmetic processing.

[0071] The condition determination portion **23** determines image processing conditions based on the asperity information identified by the asperity identification portion **22**. The image processing conditions here are conditions relating to image formation or image reading. That is, various image processing conditions including an image forming condition relating to image formation and/or an image reading condition relating to image reading executed in the image processing apparatus **10** are determined by the condition determination portion **23**. Specifically, the image processing conditions include, for example, the fixing pressure, the fixing temperature, the conveying speed of the sheet **Sh1**, the transfer voltage, the ink discharge amount in the inkjet method, or the like in the image forming portion **13**, as well as the sheet conveying speed, the light intensity, the resolution, or the like in the image reading portion **12**. For example, when the arithmetic average height (**Sa**) of the surface **A1** of the sheet **Sh1** is larger (i.e., rougher), heat may be more difficult to be transferred at the time of fixing by the image forming portion **13**, or the electrical contact resistance may be higher at the time of transferring, so that the current may be more difficult to flow. Therefore, the condition determination portion **23** automatically sets the image processing conditions based on the asperity information so as to increase the fixing temperature, decrease the conveying speed, or increase the transfer voltage when the arithmetic average height (**Sa**) becomes larger (i.e., rougher). This enables image formation and/or image reading under appropriate image

processing conditions according to the asperities on the surface A1 of the sheet Sh1, leading to an improvement in the quality (including image quality) of image formation and/or image reading. [0072] In addition, in the present embodiment, the condition determination portion 23 determines image processing conditions relating to image formation or image reading based on the fiber direction. That is, in the present embodiment, the fiber direction of the surface A1 of the sheet Sh1 is identified by the direction identification portion 24. Therefore, the condition determination portion 23 determines the image processing conditions based on not only the asperity information but also the fiber direction. For example, in the inkjet type image forming portion 13, the curl behavior differs depending on the fiber direction, so that the curl direction may be predicted in accordance with the fiber direction for curl correction. The image processing conditions determined by the condition determination portion 23 based on the fiber direction include a condition for curl correction. In addition, the “skew” in which the long side or the short side of the sheet Sh1 is tilted with respect to the conveying direction can also be estimated from the fiber direction; therefore, the image processing conditions determined by the condition determination portion 23 based on the fiber direction may include a condition for skew correction. This enables image formation and/or image reading under appropriate image processing conditions according to the fiber direction on the surface A1 of the sheet Sh1, leading to an improvement in the quality (including image quality) of image formation and/or image reading.

[0073] However, the condition determination portion 23 only has to have a function of determining the image processing conditions based on at least one of the asperity information and the fiber direction. That is, the condition determination portion 23 is not necessarily be configured to determine the image processing conditions based on both the asperity information and the fiber direction, and may determine the image processing conditions based on only one of the asperity information and the fiber direction. Further, in the present embodiment, the thickness identification portion 25 identifies the thickness of the sheet Sh1. Therefore, the condition determination portion 23 may determine the image processing conditions based on the thickness of the sheet Sh1 in addition to or instead of at least one of the asperity information and the fiber direction.

[0074] The direction identification portion 24 identifies the fiber direction of the surface A1 of the sheet Sh1 based on the identification image Im1. Here, the direction identification portion 24 identifies the fiber direction based on the deformation, distortion, or the like of the pattern light P1 in the identification image Im1. That is, depending on the line width of the pattern light P1 on the identification region R1, the degree of deformation, distortion, or the like of the pattern light P1 caused by the asperities on the surface A1 varies in accordance with the relationship between the extending direction of the pattern light P1 and the fiber direction. Therefore, in the present embodiment, the direction identification portion 24 identifies the fiber direction based at least on the variation in the line width of the pattern light P1 on the identification region R1. Thus, the fiber direction of the surface A1 of the sheet Sh1 can be identified by relatively simple arithmetic processing.

[0075] The thickness identification portion 25 identifies the thickness of the sheet Sh1 based on the output of the thickness sensor 5. That is, the thickness identification portion 25 receives an electric signal representing a physical quantity relating to the thickness of the sheet Sh1 from the thickness sensor 5, and calculates the thickness of the sheet Sh1. Since the sheet identification device 2 according to the present embodiment includes the thickness identification portion 25, it can estimate the type (paper type) of the sheet Sh1 based on not only the state of the surface A1 of the sheet Sh1 but also the thickness thereof.

[4] Sheet Identification Method

[0076] Next, a sheet identification method according to the present embodiment, i.e., the operation of the sheet identification device 2 will be described with reference to FIG. 5 to FIG. 7.

[4.1] Principle

[0077] First, the principle of the asperity identification portion 22 identifying the asperity

information based on the identification image Im1 will be described with reference to FIG. 5 and FIG. 6. In FIG. 5, the bright portion L1 of the pattern light P1 is schematically indicated by a dotted line, and the dark portion L2 is schematically indicated by a dash-dot-dot-dash line.

[0078] As shown as “CONVEX PORTION 1” in the upper part of FIG. 5, it is assumed that there is a rectangular parallelepiped convex portion A11 having a height ΔZ from the surface A1 on the surface A1 of the sheet Sh1. Here, the pattern light P1 is obliquely incident on the surface A1 of the sheet Sh1 at the predetermined angle $\theta 1$. Therefore, the pattern light P1 is projected on the same plane as the surface A1 except at the convex portion A11, and projected forward by the height ΔZ at the convex portion A11 than on the surface A1, so that the projection position of the pattern light P1 is shifted only at the convex portion A11 when viewed from the perpendicular direction of the surface A1. That is, the pattern light P1 (the bright portion L1 and the dark portion L2) of a portion provided with the convex portion A11 in the identification image Im1 captured by the imaging portion 4 is shifted from the original projection position on the surface A1 by the shift amount ΔX represented by the following Equation 1:

[00001]
$$X = Z / \tan \theta 1 \quad (\text{Equation 1})$$

[0079] Since the predetermined angle $\theta 1$ is known, when the shift amount ΔX is obtained from the identification image Im1, the height ΔZ of the convex portion A11 can be calculated from the shift amount ΔX and the Equation 1. Then, the asperity information of the entire identification region R1 can be obtained from the shift amounts ΔX of the entire identification region R1. The asperity information calculated in this way has a correlation with the arithmetic average height (Sa) of the surface A1.

[0080] In addition, as shown as “CONVEX PORTION 2” in the lower part of FIG. 5, even when there is a triangular prism-shaped convex portion A12 whose height ΔZ from the surface A1 varies from part to part on the surface A1 of the sheet Sh1, the heights ΔZ of the respective parts of the convex portion A12 can be calculated. That is, similarly to the above example, by obtaining the shift amounts ΔX from the identification image Im1, the heights ΔZ of the convex portion A12 can be calculated from the above Equation 1. Therefore, for example, as in the case of the paper sheet Sh1, the asperity information can also be calculated for the asperities caused by the undulation component generated by continuous entanglement of many fibers.

[0081] By the way, in the method for obtaining the roughness of the surface A1 from the shadow image caused by the asperities as in the above-described related technique, for example, in the case of the paper sheet Sh1, the local fiber asperities are strongly reflected in the calculation result, so that the calculation result does not necessarily have a linear relationship with the arithmetic average height (Sa). Therefore, in the above-described method of the related technique, it is difficult to determine from the calculation result the magnitude of the surface roughness of the sheet Sh1 of the same type (for example, plain paper), although it may be possible to discriminate between glossy paper with a high flatness (gloss paper) and plain paper, for example. Therefore, in the above-described method of the related technique, in order to determine the magnitude of the surface roughness, it is necessary to prepare in advance, for example, a table (database) in which the calculation results of various sheets Sh1 are associated with arithmetic average heights (Sa).

[0082] In contrast, in the sheet identification device 2 according to the present embodiment, by optimizing the line width, the predetermined angle $\theta 1$, and the like of the pattern light P1, the asperity information having a high linearity with the arithmetic average height (Sa) can be calculated while also reducing the influence of local fibers. Therefore, according to the method of the present embodiment, it is possible to uniquely obtain an arithmetic average height (Sa) from the calculation result of the asperity identification portion 22 without preparing in advance the table (database) in which the calculation results (asperity information) are associated with arithmetic average heights (Sa).

[0083] FIG. 6 shows an example of the identification image Im1 obtained by the sheet

identification device 2 according to the present embodiment. Here, as the imaging conditions of the identification image Im1, it is assumed that the predetermined angle $\theta 1$ is 40 degrees, the resolution (number of pixels) of the imaging element 41 is 100×100 , the line width W1 (see FIG. 4) of the bright portion L1 of the stripe pattern produced by the pattern light P1 is $120 \mu\text{m}$, and the line width W2 (see FIG. 4) of the dark portion L2 is $120 \mu\text{m}$. Further, it is assumed that the image magnification M is “1”, and the irradiation direction of the pattern light P1, that is, the arrangement direction of the bright portion L1 and the dark portion L2 is the same as the fiber direction.

[0084] The identification image Im1 is composed of a plurality of pixels, and each of the plurality of pixels has a pixel value corresponding to luminance. In the present embodiment, as an example, the relationship between the luminance and the pixel value is defined such that the higher the luminance, the larger the pixel value. Therefore, in the identification image Im1 obtained by capturing the identification region R1 on which the pattern light P1 is projected, the pixel values of the pixels corresponding to the bright portion L1 are relatively large values, and the pixel values of the pixels corresponding to the dark portion L2 are relatively small values.

[0085] The upper part (“Sa: SMALL”) of FIG. 6 shows a identification image Im1 when the pattern light P1 is projected on an identification region R1 of glossy paper (gloss paper) having a small arithmetic average height (Sa), that is, having a high flatness. On the other hand, the lower part (“Sa: LARGE”) of FIG. 6 shows an identification image Im1 when the pattern light P1 is projected on an identification region R1 of plain paper (Sa= $6 \mu\text{m}$) having a large arithmetic average height (Sa), that is, having a low flatness. As shown in FIG. 6, as the arithmetic average height (Sa) increases, the deformation and distortion of the stripe pattern produced by the pattern light P1 in the identification image Im1 increase. Specifically, when the deformation and distortion of the stripe pattern increase, the boundary line between the bright portion L1 and the dark portion L2 of the stripe pattern is distorted, and the variation in the line width of each of the bright portion L1 and the dark portion L2 increases. That is, as the arithmetic average height (Sa) increases, the undulation component of the height of the surface A1 increases, so that the variation in the line width of each of the bright portion L1 and the dark portion L2 increases. As described above, in FIG. 6, it is apparent that, deformation, distortion, or the like occurs in the stripe pattern produced by the pattern light P1 in the identification image Im1, in accordance with the asperities on the surface A1.

[4.2] Specific Processing

[0086] Next, specific processing for identifying the asperity information based on the identification image Im1 by the asperity identification portion 22 will be described with reference to FIG. 7. Steps S1, S2, . . . in the flowchart shown in FIG. 7 represent the numbers of the processing procedure (steps) executed by the control portion 16. The processing to be described below is started in accordance with the timing at which the sheet Sh1 passes the position (monitor position) corresponding to the sensor unit 20 of the conveying path T1, for example.

[0087] When the purpose is to restore a three-dimensional shape, the analysis of the identification image Im1 including the pattern light P1 can be realized by, for example, a method of continuously projecting a plurality of pattern lights P1 and using a Fourier transform or the like of the identification images Im1 to calculate a phase change of the pattern light P1. However, this method has a relatively high calculation load, takes a relatively long time to calculate the roughness (asperity information) of the surface A1, and also requires relatively high hardware (CPU, GPU, memory, etc.) costs. Therefore, in the present embodiment, instead of the above-described method, the following method is adopted so that the roughness (asperity information) of the surface A1 can be calculated by relatively simple arithmetic processing.

[0088] That is, in the present embodiment, the asperity identification portion 22 calculates the width (line width) of at least one of the bright portion L1 and the dark portion L2 for each row (each line) of the identification image Im1, with the arrangement direction (left-right direction in FIG. 6) of the bright portion L1 and the dark portion L2 in the identification image Im1 as a “row

direction”. That is, the identification image **Im1** is an “N pixels×M rows” image in which N pixels are arranged in the arrangement direction of the bright portion **L1** and the dark portion **L2**, and the asperity identification portion **22** calculates the line width of each row of the M rows included in the identification image **Im1**. The asperity identification portion **22** obtains the line width of at least one of the bright portion **L1** and the dark portion **L2** calculated for each row as described above for the entire identification image **Im1**, and calculates the magnitude of the variation in the line width in the identification image **Im1** as the asperity information. Thus, the asperity identification portion **22** can obtain asperity information having a correlation with the arithmetic average height (**Sa**) of the surface **A1**, based on the variation in the line width of the pattern light **P1**.

Step S1

[0089] Specifically, in step **S1**, the control portion **16** determines whether the sheet **Sh1** has reached the monitor position, that is, the position corresponding to the sensor unit **20** of the conveying path **T1**. When the sheet feed portion **14** supplies the sheet **Sh1** to the image forming portion **13**, the control portion **16** determines that the sheet **Sh1** has reached the monitor position when the sheet **Sh1** is detected by a sensor at the monitor position (**S1**: Yes), and shifts the processing to step **S2**. On the other hand, if the sheet **Sh1** is not detected by the sensor at the monitor position, the control portion **16** determines that the sheet **Sh1** has not reached the monitor position (**S1**: No), and shifts the processing to step **S1**.

Steps S2 and S3

[0090] In step **S2**, the control portion **16** controls, at the acquisition portion **21**, the light irradiation portion **3** to cause the light irradiation portion **3** to apply the pattern light **P1**. Thus, the pattern light **P1** is projected on the identification region **R1** of the surface **A1** of the sheet **Sh1**. In step **S3**, the control portion **16** controls, at the acquisition portion **21**, the imaging portion **4** to cause the imaging portion **4** to image the identification region **R1** on which the pattern light **P1** is being projected. Thus, an identification image **Im1**, which is an image of the identification region **R1** of the surface **A1** of the sheet **Sh1**, is generated by the imaging portion **4**.

Step S4

[0091] In step **S4**, the control portion **16** acquires, at the acquisition portion **21**, an image of one row (one line) of the identification image **Im1** from the imaging portion **4**. That is, the acquisition portion **21** acquires one row of the identification image **Im1** corresponding to one pixel in the column direction. Since the imaging portion **4** (imaging element **41**) is generally designed to sequentially read out an image for each row, the amount of memory used can be kept low by acquiring and analyzing (steps **S5** and **S6**) the identification image **Im1** for each row in this manner.

Step S5

[0092] In step **S5**, the control portion **16** executes, at the acquisition portion **21**, preprocessing on the identification image **Im1**. At this time, one row (one line) of the identification image **Im1** acquired in step **S4** is subjected to the preprocessing. That is, the control portion **16** executes the preprocessing on the identification image **Im1** row by row. The preprocessing includes, for example, filtering processing and binarization processing. Specifically, the control portion **16** performs noise removal or the like in the filtering processing and further performs binarization with a reference value for one row of the identification image **Im1**.

[0093] The reference value used in the binarization processing is, for example, an average value of a plurality of pixels, a value determined in advance (predetermined value), or the like. The pixels corresponding to the bright portion **L1** become “white pixels” as pixels having pixel values equal to or greater than the reference value, and the pixels corresponding to the dark portion **L2** become “black pixels” as pixels having pixel values less than the reference value. The preprocessing may include trimming processing for cutting out only a part of the identification image **Im1** to narrow down the area to be processed in step **S6**. In addition, the filtering processing and the like are not essential, and may be omitted as appropriate.

Step S6

[0094] In step S6, the control portion **16** extracts, at the asperity identification portion **22**, width data indicating the width (line width) of at least one of the bright portion **L1** and the dark portion **L2** from the identification image **Im1**. At this time, the width data is extracted from one row (one line) of the identification image **Im1** acquired in step S4. That is, the control portion **16** executes extraction of width data on the identification image **Im1** row by row. Specifically, the control portion **16** calculates, as the width data, the number of white pixels corresponding to the bright portion **L1** and the number of black pixels corresponding to the dark portion **L2** in one row of the identification image **Im1**. At this time, the control portion **16** extracts the number of white pixels and the number of black pixels throughout one row of the identification image **Im1**, thereby extracting the sum of the line widths of the plurality of bright portions **L1** and the sum of the line widths of the plurality of dark portions **L2**.

[0095] In the present embodiment, as an example, both the number of white pixels corresponding to the line widths of the bright portions **L1** and the number of black pixels corresponding to the line widths of the dark portions **L2** are used as the width data, but the present disclosure is not limited to this example, and only the number of pixels of either the bright portions **L1** or the dark portions **L2** may be used as the width data. That is, the control portion **16** may identify the asperity information by focusing on the line widths of either the bright portions **L1** or the dark portions **L2**. In addition, the control portion **16** may extract the line width of each bright portion **L1** and that of each dark portion **L2** by extracting the number of white pixels consecutive in the row direction and the number of black pixels consecutive in the row direction. In this case, the control portion **16** may use the line width of each of the plurality of bright portions **L1** (or dark portions **L2**) as the width data, or may use a representative value (for example, an average value, a mode value, a median value, or the like) of the line widths of the plurality of bright portions **L1** (or dark portions **L2**) as the width data.

Step S7

[0096] In step S7, the control portion **16** determines whether or not processing has been completed to the last one row of the identification image **Im1**. That is, with respect to the “N pixels×M rows” identification image **Im1**, if the processing target is the M-th row which is the last row, the control portion **16** determines that the processing has been completed to the last row (S7: Yes), and shifts the processing to step S8. On the other hand, if the processing target is not the M-th row which is the last row, the control portion **16** determines that the processing has not been completed to the last row (S7: No), and shifts the processing to step S4 to acquire the next one row of the identification image **Im1**.

Step S8

[0097] In step S8, the control portion **16** calculates, at the asperity identification portion **22**, the standard deviation σ of the width data of the M rows of the identification image **Im1**. As the arithmetic average height (S_a) increases, the undulation component of the height of the surface **A1** increases, so that the variation in the line width of each of the bright portion **L1** and the dark portion **L2** increases (see FIG. 6), and the standard deviation σ increases. That is, the asperity identification portion **22** calculates the standard deviation σ as the asperity information.

Step S9

[0098] In step S9, the control portion **16** determines, at the condition determination portion **23**, image processing conditions. That is, the condition determination portion **23** determines the image processing conditions including image forming conditions in accordance with the standard deviation σ calculated in step S8. As an example, when the standard deviation σ increases, the condition determination portion **23** sets the image forming conditions so as to increase the fixing temperature, decrease the conveying speed, or increase the transfer voltage. Thus, when an image is formed on the sheet **Sh1** by the image forming portion **13**, the image forming conditions corresponding to the asperities on the surface **A1** of the sheet **Sh1** are automatically applied.

[0099] The procedure of the sheet identification method described above is merely an example, and

the order of the processes shown in the flowchart of FIG. 7 may be changed as appropriate.

[5] Irradiation Angle

[0100] Next, the irradiation angle of the pattern light P1 will be described with reference to FIG. 8, which shows examples of the identification image Im1 when the predetermined angle $\theta 1$ is changed. In FIG. 8, as the imaging conditions of the identification image Im1, it is assumed that the arithmetic average height (Sa) of the identification region R1 is 6 μm , the resolution (number of pixels) of the imaging element 41 is 100×100, the line width W1 of the bright portion L1 of the stripe pattern produced by the pattern light P1 is 100 μm , and the line width W2 of the dark portion L2 is 100 μm . Further, it is assumed that the irradiation direction of the pattern light P1, that is, the arrangement direction of the bright portion L1 and the dark portion L2 is the same as the fiber direction.

[0101] The optical axis Ax1 of the pattern light P1 is inclined at the predetermined angle $\theta 1$ with respect to the surface A1 of the sheet Sh1 (see FIG. 3). Here, the predetermined angle $\theta 1$ has a great influence on the brightness of the identification image Im1. In the method for obtaining the roughness of the surface A1 from the shadow image caused by the asperities as in the related technique described above, the irradiation angle (corresponding to the predetermined angle $\theta 1$) of the light with respect to the surface A1 of the sheet Sh1 is set relatively shallow (small) in order to capture the asperities on the order of several micrometers as a shadow image. In particular, in order to increase the sensitivity to asperities, the irradiation angle is set at a very shallow angle of about 10 degrees. However, at very small irradiation angles, sufficient light does not reach the imaging portion 4, and the identification image Im1 becomes a relatively dark image, so that a relatively expensive high-sensitivity imaging element 41 is required to obtain the roughness of the surface A1 from the dark image.

[0102] In contrast, in the sheet identification device 2 according to the present embodiment, the roughness of the surface A1 is obtained from the degree of deformation, distortion, or the like of the pattern light P1 in the identification image Im1; therefore, it is sufficient that deformation, distortion, or the like of the pattern light P1 is caused by asperities. Therefore, in the present embodiment, the predetermined angle $\theta 1$ can be set larger than in the method of the related technique described above, and a bright image can be realized as the identification image Im1. Therefore, the roughness of the surface A1 can be obtained from the identification image Im1 even with a relatively inexpensive imaging element 41.

[0103] Rather, in the configuration of the present embodiment, as shown in FIG. 8, when the predetermined angle $\theta 1$ becomes small, the shape of the pattern light P1 projected on the identification region R1 collapses, and the boundary between the bright portion L1 and the dark portion L2 of the stripe pattern becomes ambiguous. That is, as is clear from FIG. 8, the shape of the pattern light P1 collapses more when the predetermined angle $\theta 1$ is 30 degrees than when it is 40 degrees, when the predetermined angle $\theta 1$ becomes 20 degrees, the shape of the pattern light P1 collapses further, and when the predetermined angle $\theta 1$ becomes 10 degrees, the shape of the pattern light P1 collapses even further. From the identification image Im1 of the pattern light P1 whose shape has been collapsed as described above, it is difficult to extract deformation, distortion, or the like of the pattern light P1 caused by asperities. The inventors have verified various predetermined angles $\theta 1$, and have found that the predetermined angle $\theta 1$ is preferably 20 degrees or more when the sheet Sh1 having an arithmetic average height (Sa) of the surface A1 of about several micrometers is the target.

[0104] In short, in consideration of the brightness of the identification image Im1, the predetermined angle $\theta 1$ is preferably 10 degrees or more, and more preferably 15 degrees or more. Further, in the present embodiment, the predetermined angle $\theta 1$ is set at 20 degrees or more so that the shape of the pattern light P1 is not too deformed. That is, the predetermined angle $\theta 1$ is 20 degrees or more and 90 degrees or less. Here, the lower limit value of the predetermined angle $\theta 1$ is not limited to 20 degrees, and may be, for example, 25 degrees, 30 degrees, 35 degrees, 40 degrees,

45 degrees, 50 degrees, 55 degrees, 60 degrees, 65 degrees, 70 degrees, 75 degrees, or 80 degrees. Also, the upper limit value of the predetermined angle $\theta 1$ is not limited to 90 degrees, and may be, for example, 85 degrees, 80 degrees, 75 degrees, 70 degrees, 65 degrees, 60 degrees, 55 degrees, 50 degrees, or 45 degrees.

[0105] Since the difference between 90 degrees and the predetermined angle $\theta 1$ corresponds to the “incident angle” which is the angle between the pattern light P1 and the perpendicular line of the surface A1, the “incident angle” of the pattern light P1 when the predetermined angle $\theta 1$ is 20 degrees is 70 degrees (=90 degrees–20 degrees). On the other hand, when the predetermined angle is 90 degrees, the incident angle of the pattern light P1 is 0 degrees.

[6] Line Width

[0106] Next, the line width of the pattern light P1 will be described later with reference to FIG. 9 and FIG. 10. FIG. 9 is a graph showing the relationship between the standard deviation σ as the asperity information calculated in accordance with the flowchart of FIG. 7 and the actual arithmetic average height Sa for 48 types of sheets Sh1. In FIG. 9, the horizontal axis represents the arithmetic average height Sa, and the vertical axis represents the standard deviation σ (the standard deviation σ is an average value of five locations in the bright portion L1 or the dark portion L2). In FIG. 9, as the imaging conditions of the identification image Im1, it is assumed that the resolution (number of pixels) of the imaging element 41 is 100×100, the line width W1 of the bright portion L1 of the stripe pattern produced by the pattern light P1 is 80 μm , and the line width W2 of the dark portion L2 is 80 μm . Further, it is assumed that the irradiation direction of the pattern light P1, that is, the arrangement direction of the bright portion L1 and the dark portion L2 is the same as the fiber direction.

[0107] In the example of FIG. 9, the determination coefficient R.sup.2 of the linear regression model of the arithmetic average height Sa and the standard deviation σ is “0.9684”, which confirms a high linearity between the standard deviation σ as the asperity information and the arithmetic average height Sa.

[0108] On the other hand, FIG. 10 shows the results of calculating the determination coefficient R.sup.2 while changing the line width of the pattern light P1 and the relationship between the irradiation direction of the pattern light P1 and the fiber direction. In FIG. 10, as the imaging conditions of the identification image Im1, it is assumed that the predetermined angle $\theta 1$ is 40 degrees and the resolution (number of pixels) of the imaging element 41 is 100×100. Under the conditions, the line width W1 of the bright portion L1 and the line width W2 of the dark portion L2 of the stripe pattern produced by the pattern light P1 are changed in the range of 40 μm to 200 μm , and the relationship between the irradiation direction of the pattern light P1 and the fiber direction is changed between “the same” (that is, parallel) and “90°” (that is, orthogonal).

[0109] According to FIG. 10, it is presumed that the influence of the relationship between the irradiation direction of the pattern light P1 and the fiber direction on the standard deviation σ as the asperity information varies depending on the line width of the pattern light P1. That is, in FIG. 10, it can be confirmed that the smaller (thinner) the line width of the pattern light P1, the greater the influence of the relationship between the irradiation direction of the pattern light P1 and the fiber direction on the standard deviation σ as the asperity information. It is considered that this is because the width of the fiber of the sheet Sh1 is several tens of micrometers or less, and the line width of the pattern light P1 approaches the fiber width, which makes the local undulation component of the height of the surface A1 likely to occur as the variation in the line width of each of the bright portion L1 and the dark portion L2.

[0110] When the line width of the pattern light P1 is 100 μm or more, the determination coefficient R.sup.2 is “0.85” or more regardless of the relationship between the irradiation direction of the pattern light P1 and the fiber direction. Therefore, when the line width of the pattern light P1 is 100 μm or more, the influence of the relationship between the irradiation direction of the pattern light P1 and the fiber direction on the standard deviation σ as the asperity information is relatively small,

and the relationship between the irradiation direction of the pattern light P1 and the fiber direction is negligible. In short, whether the relationship between the irradiation direction of the pattern light P1 and the fiber direction affects the asperity information is determined depending on whether the line width W1 of the bright portion L1 and the line width W2 of the dark portion L2 of the stripe pattern produced by the pattern light P1 are 100 μm or more, or less than 100 μm . Namely, when the line width is 100 μm or more, the relationship between the irradiation direction and the fiber direction hardly affects the asperity information, so that the calculated asperity information can be regarded as “independent of the fiber direction”. On the other hand, when the line width is less than 100 μm , the relationship between the irradiation direction and the fiber direction is likely to affect the asperity information, so that the calculated asperity information can be regarded as “dependent on the fiber direction”.

[0111] As described above, even in the sheet identification device 2 according to the present embodiment, the relationship between the irradiation direction of the pattern light P1 and the fiber direction may affect the standard deviation σ as the asperity information, depending on the line width of the pattern light P1. By making the line width of the pattern light P1 relatively larger than the width of the fiber of the sheet Sh1, the influence can be reduced, which causes a high linearity between the standard deviation σ as the asperity information and the arithmetic average height Sa.

[0112] From the above, in the present embodiment, the width of at least one of the bright portion L1 and the dark portion L2 is preferably 60 μm or more and 500 μm or less. Furthermore, in order to make the asperity information less susceptible to the relationship between the irradiation direction of the pattern light P1 and the fiber direction, it is preferable that at least one of the line width W1 of the bright portion L1 and the line width W2 of the dark portion L2 of the stripe pattern produced by the pattern light P1 is 100 μm or more. Conversely, in order to make the asperity information more susceptible to the relationship between the irradiation direction of the pattern light P1 and the fiber direction, it is preferable that at least one of the line width W1 of the bright portion L1 and the line width W2 of the dark portion L2 of the stripe pattern produced by the pattern light P1 is less than 100 μm . Here, the lower limit value of the width of at least one of the bright portion L1 and the dark portion L2 is not limited to 60 μm , and may be, for example, 65 μm , 70 μm , 75 μm , 80 μm , 85 μm , 90 μm , or 95 μm . Also, the upper limit value of the width of at least one of the bright portion L1 and the dark portion L2 is not limited to 500 μm , and may be, for example, 450 μm , 400 μm , 350 μm , 300 μm , 250 μm , 200 μm , 180 μm , 160 μm , 140 μm , or 120 μm .

[7] Lattice Pattern

[0113] Next, the pattern light P1 that produces a lattice pattern will be described with reference to FIG. 11. Namely, the projection of the pattern light P1 on the identification region R1 generates a luminance distribution of a lattice pattern including the bright portion L1 and the dark portion L2.

[0114] The lattice pattern is a superposition of a vertical stripe pattern and a horizontal stripe pattern, which are orthogonal to each other. Therefore, as shown in FIG. 11, the pattern light P1 that produces the lattice pattern can be regarded as combined light of first pattern light P11 that produces a first stripe pattern (vertical stripe pattern) and second pattern light P12 that produces a second stripe pattern (horizontal stripe pattern). In the example of FIG. 11, the dark portion L2 is prioritized in portions where the bright portion L1 and the dark portion L2 of the first pattern light P11 and the second pattern light P12 overlap, but the present disclosure is not limited to this, and the bright portion L1 may be prioritized. That is, the bright portion L1 (shaded) and the dark portion L2 (blackened) of the lattice pattern shown in FIG. 11 may be reversed.

[0115] Here, the line width of the first pattern light P11 and the line width of the second pattern light P12 are different from each other. That is, the vertical stripe and the horizontal stripe of the lattice pattern have different line widths. In the example of FIG. 11, the line width W11 of the bright portion L1 and the line width W12 of the dark portion L2 of the first pattern light P11 are larger than the line width W21 of the bright portion L1 and the line width W22 of the dark portion

L2 of the second pattern light P12. Specifically, the line width W11 of the bright portion L1 and the line width W12 of the dark portion L2 of the first pattern light P11 are both 100 μm , and the line width W21 of the bright portion L1 and the line width W22 of the dark portion L2 of the second pattern light P12 are both 80 μm . Therefore, in the example of FIG. 11, of the lattice pattern, the vertical stripe pattern is wider than the horizontal stripe pattern.

[0116] In short, the stripe patterns include a first stripe pattern (vertical stripe pattern) and a second stripe pattern (horizontal stripe pattern) which are orthogonal to each other. The width of at least one of the bright portion L1 and the dark portion L2 is different between the first stripe pattern and the second stripe pattern. This makes it possible to identify the fiber direction of the sheet Sh1 in addition to the asperity information. In the example of FIG. 11, the first stripe pattern is wider than the second stripe pattern for both the bright portion L1 and the dark portion L2, but the present disclosure is not limited to this example, and the first stripe pattern may be wider than the second stripe pattern for only the bright portion L1 or only the dark portion L2. In addition, the second stripe pattern may be wider than the first stripe pattern for at least one of the bright portion L1 and the dark portion L2.

[0117] In addition, the pattern light P1 that produces the above-described lattice pattern may be realized using a lattice-shaped shield 32 or two shields 32 in which slits 321 are formed. In the latter case, the lattice pattern as shown in FIG. 11 can be realized by stacking two shields 32 having slits 321 with different widths such that the orientations of their slits 321 are orthogonal to each other.

[0118] When the pattern light P1 of such a lattice pattern is used, the control portion 16 can analyze the line width of the second stripe pattern (horizontal stripe pattern) in addition to the line width of the first stripe pattern (vertical stripe pattern) when analyzing the identification image Im1. That is, the control portion 16 can calculate the variation in the line width in the horizontal direction from the first stripe pattern, and can calculate the variation in the line width in the vertical direction from the second stripe pattern. In this way, it is possible to acquire at one time the identification image Im1 required for the analysis in the two directions, i.e., the vertical direction and the horizontal direction, orthogonal to each other. In this case, since the identification image Im1 cannot be acquired and analyzed for each row, the entire identification image Im1 needs to be stored in the memory; however, comparison of the calculation result for the vertical direction with the calculation result for the horizontal direction enables identification of the fiber direction.

[0119] That is, the control portion 16 can identify the fiber direction of the surface A1 of the sheet Sh1 by the direction identification portion 24 based on the difference between the calculation result for the vertical direction and the calculation result for the horizontal direction. In the example of FIG. 11, since the line widths W11 and W12 of the first pattern light P11, which projects the first stripe pattern (vertical stripe pattern), are 100 μm or more, the asperity information for the horizontal direction calculated from the first pattern light P11 is regarded as “independent of the fiber direction”. On the other hand, since the line widths W21 and W22 of the second pattern light P12, which projects the second stripe pattern (horizontal stripe pattern), are less than 100 μm , the asperity information for the vertical direction calculated from the second pattern light P12 is regarded as “dependent on the fiber direction”.

[0120] Therefore, when the same result (asperity information) is obtained for the vertical direction and the horizontal direction, the direction identification portion 24 determines that the fiber direction is orthogonal to the second stripe pattern (horizontal stripe pattern) produced by the second pattern light P12. In other words, it is determined that the fiber direction is the same as the arrangement direction of the bright portion L1 and the dark portion L2 of the second pattern light P12. On the other hand, when different results (asperity information) are obtained for the vertical direction and the horizontal direction, the direction identification portion 24 determines that the fiber direction is along the second stripe pattern (horizontal stripe pattern) produced by the second pattern light P12. In other words, it is determined that the fiber direction is orthogonal to the

arrangement direction of the bright portion **L1** and the dark portion **L2** of the second pattern light **P12**. Here, whether the asperity information is the same for the vertical direction and the horizontal direction is determined by whether the difference between the asperity information for the vertical direction and the asperity information for the horizontal direction is less than or equal to a predetermined value, and when the difference is less than or equal to the predetermined value, it is determined that the asperity information is the same for the vertical direction and the horizontal direction.

[0121] In this way, by narrowing the line width of only one of the stripe patterns of the lattice pattern to obtain the asperity information “dependent on the fiber direction”, both the asperity information and the fiber direction can be identified from the identification image **Im1**. When the fiber direction is identified, for example, the inkjet type image forming portion **13** can predict the curl direction in accordance with the fiber direction, and the condition determination portion **23** can perform curl correction.

[8] Modifications

[0122] The plurality of constituent elements included in the image processing apparatus **10** may be distributed across a plurality of housings. For example, at least one of the acquisition portion **21**, the asperity identification portion **22**, the condition determination portion **23**, the direction identification portion **24**, the thickness identification portion **25**, and the like, which are constituent elements of the sheet identification device **2**, is not necessarily realized as a function of the control portion **16**, and may be provided in a separate housing from the control portion **16**. That is, the sheet identification device **2** need not necessarily be integrated with the image processing apparatus **10**, and at least a part of the sheet identification device **2** may be provided in a separate housing from the image processing apparatus **10**.

[0123] In addition, the sheet identification device **2** need to have at least the function of identifying the asperity information on the asperities on the surface **A1** of the sheet **Sh1**, and the functions of identifying the fiber direction of the sheet **Sh1**, the thickness of the sheet **Sh1**, and the like may be omitted as appropriate. For example, when the function of identifying the thickness of the sheet **Sh1** is omitted, the thickness sensor **5** and the thickness identification portion **25** may be omitted.

[0124] In addition, in the first embodiment, an example is shown in which the optical axis **Ax1** of the light irradiation portion **3** is inclined at the predetermined angle $\theta 1$ with respect to the identification region **R1** of the sheet **Sh1**, and the optical axis **Ax2** of the imaging portion **4** is orthogonal to the identification region **R1** of the sheet **Sh1**, but the present disclosure is not limited to this configuration. For example, the optical axis **Ax1** of the light irradiation portion **3** may be orthogonal to the identification region **R1** of the sheet **Sh1**, the optical axis **Ax2** of the imaging portion **4** may be inclined with respect to the identification region **R1** of the sheet **Sh1**, or both the optical axis **Ax1** and the optical axis **Ax2** may be inclined with respect to the identification region **R1** of the sheet **Sh1**.

[0125] In addition, the light irradiation portion **3** may include, for example, a projector, and project any pattern light **P1** input as projection data on the identification region **R1**. That is, an image projected from the projector may be projected on the identification region **R1** as the pattern light **P1**. In this case, it is also easy to employ a moving image as the pattern light **P1**.

[0126] In addition, the sheet **Sh1** to be irradiated with the pattern light **P1** is not limited to the sheet being conveyed, and may be, for example, the sheet **Sh1** set in the sheet feed cassette **141**. In this case, by moving at least one of the sheet **Sh1** and the imaging portion **4** to capture the identification image **Im1** in a state where the sheet **Sh1** and the imaging portion **4** are relatively moved, a wide area of the sheet **Sh1** can be imaged while reducing the image magnification.

Second Embodiment

[0127] The image processing apparatus **10A** according to the present embodiment differs from the image processing apparatus **10** according to the first embodiment in that the sheet identification device **2A** includes an output portion **26** as shown in FIG. **12**. In the following, structures similar to

those of the first embodiment are denoted by common reference numerals, and descriptions thereof are omitted as appropriate.

[0128] The output portion **26** outputs the identification result of at least one of the asperity identification portion **22**, the direction identification portion **24**, and the thickness identification portion **25**. In the present embodiment, as an example, the output portion **26** outputs the identification result by causing the operation display portion **15** to display the identification result so as to notify the user of the identification result. The mode of the output of the identification result by the output portion **26** is not limited to display on the operation display portion **15**, but may be transmission to an external device, writing to a non-temporary recording medium readable by a computer system, or the like. The output portion **26** is provided in the control portion **16** as a function of the control portion **16**.

[0129] In the case of the identification result of the asperity identification portion **22**, what is output from the output portion **26** is, for example, the standard deviation σ as the asperity information, the arithmetic average height (S_a), or information representing the type of the sheet Sh1. Similarly, in the case of the identification result of the direction identification portion **24**, what is output by the output portion **26** is, for example, the fiber direction, or information indicating “vertical grain” or “horizontal grain”.

[0130] In addition, the output portion **26** may output information such as a life estimation result, a recommendation for maintenance timing, or a recommendation for the type of the sheet Sh1, which is estimated from the identification result of the asperity identification portion **22** or the like. For example, parts of the image processing apparatus **10A** may be worn when the sheet Sh1 is conveyed, but the rougher the surface A1 of the conveyed sheet Sh1, the more likely the wear progresses. That is, since the degree of deterioration of the image processing apparatus **10A** differs depending on the surface roughness or the like of the sheet Sh1 used, the accuracy of the life estimation of the image processing apparatus **10A** is improved if the asperity information of the sheet Sh1 is known in addition to the number of conveyed sheets Sh1, for example. Therefore, the output portion **26** can output information such as the life estimation result of the image processing apparatus **10A** or a recommendation for maintenance timing of the image processing apparatus **10A** by, for example, causing the operation display portion **15** to display the information so as to notify the user of the information. Further, in order to extend the life of the image processing apparatus **10A**, the output portion **26** can notify the user of information such as a recommendation for the sheet Sh1 having a higher flatness than the sheet Sh1 in use.

[0131] In particular, in the sheet identification device **2A** according to the present embodiment, as described in the first embodiment, asperity information having a high linearity with the arithmetic average height (S_a) can be calculated. Therefore, even a sheet Sh1 that is not registered in the database or the like in advance can be reflected, for example, in the life estimation of the image processing apparatus **10A**.

[0132] The output portion **26** may also output information such as the result of estimation on whether the sheet Sh1 faces up or down, which is estimated from the identification result of the asperity identification portion **22**. That is, depending on the type of sheet Sh1, the roughness may be different on the front and back sides of the sheet Sh1, such as the back side being rougher than the front side. Therefore, if the asperity information of each of the front and back sides of the sheet Sh1 is known, it is possible to estimate whether the sheet Sh1 faces up or down. Therefore, the output portion **26** can output information such as the result of estimation on whether the sheet Sh1 faces up or down by, for example, causing the operation display portion **15** to display the information so as to notify the user of the information. In this case, it is necessary to capture the identification images Im1 of both sides of the sheet Sh1 in the thickness direction. Therefore, two sensor units **20** may be disposed so as to sandwich the conveying path T1, or one sensor unit **20** may capture the identification images Im1 of both sides using a mirror or the like, or the sheet Sh1 may be turned over.

[0133] As a modification of the second embodiment, the condition determination portion **23** may be omitted as appropriate.

Third Embodiment

[0134] In the present embodiment, the asperity identification portion **22** identifies the asperity information on the asperities on the surface **A1** of the sheet **Sh1** based on an integrated image **Im10**, instead of using the identification image **Im1** as it is. The integrated image **Im10** is an image obtained by integrating the identification image **Im1** when the identification region **R1** moves by a certain amount of movement within the surface **A1** of the sheet **Sh1**. As described above, by identifying the asperity information using the integrated image **Im10** obtained by integrating the identification image **Im1** instead of using the identification image **Im1** as it is, the asperity information can be more easily identified even if there is “blurring” in the identification image **Im1**. That is, in the present embodiment, since the asperity information is identified based on the integrated image **Im10** which can include a “blurring” component caused by integration in the first place, the asperity information can be identified regardless of “blurring” in the identification image **Im1**, and the accuracy of identifying the asperities can be easily improved as compared with the related technique. In the following, structures similar to those of the first embodiment are denoted by common reference numerals, and descriptions thereof are omitted as appropriate.

[0135] The “blurring” of an image such as the identification image **Im1** in the present disclosure means that the subject in the obtained image appears as multiple layers due to the relative movement of the subject and the imaging portion **4** during imaging of the subject by the imaging portion **4**, resulting in an unclear image. For example, if a high-speed camera or the like is used as the imaging portion **4**, such “blurring” is less likely to occur, but this type of imaging portion **4** tends to be expensive.

[0136] The “image integration” in the present disclosure means integration of the pixel value (luminance value) for each pixel, and is realized, for example, by continuously integrating the pixel value for 10 ms for each pixel when the exposure time of the imaging portion **4** which captures the identification image **Im1** is 10 ms. That is, by using the time required for the identification region **R1** to move by a certain amount of movement as the exposure time and continuously integrating the pixel value during the exposure time, the integrated image **Im10**, which is an integration of the identification image **Im1** when the identification region **R1** moves by a certain amount of movement within the surface **A1** of the sheet **Sh1**, is obtained. The “image integration” includes not only such continuous integration of the pixel value, but also discontinuous integration such as integration (addition) of pixel values of a plurality of identification images **Im1** for each pixel. That is, by integrating the pixel values of a plurality of identification images **Im1** intermittently obtained while the identification region **R1** moves by a certain amount of movement, an integrated image **Im10**, which is an integration of the identification images **Im1** obtained while the identification region **R1** moves by a certain amount of movement within the surface **A1** of the sheet **Sh1**, is obtained. Further, the average value of pixel values for a predetermined number of pixels can be obtained by integrating the pixel values for the predetermined number of pixels and then dividing the integrated pixel value by the predetermined number of pixels. Such averaging processing for obtaining an average value is also included in the “image integration”.

[0137] In the present embodiment, as an example, the pattern light **P1** forms a stripe pattern in which a bright portion **L1** and a dark portion **L2** are alternately arranged on the identification region **R1**, as shown in FIG. **13**. That is, the projection of the pattern light **P1** produces a luminance distribution of a stripe pattern including the bright portions **L1** and the dark portions **L2** on the identification region **R1**. Here, the bright portions **L1** are regions brighter than the dark portions **L2**, and in other words, the stripe pattern is a pattern in which a plurality of bright portions **L1** are arranged at intervals or a pattern in which a plurality of dark portions **L2** are arranged at intervals. That is, in the identification region **R1**, the pattern light **P1** is projected to form a stripe pattern in which a linear bright portion **L1** and a linear dark portion **L2** are alternately arranged in a direction

orthogonal to the longitudinal directions of the linear bright portion **L1** and the linear dark portion **L2**. In the present embodiment, as an example, a linear bright portion **L1** and a linear dark portion **L2**, which extend along the conveying direction **D1** of the sheet **Sh1**, are arranged alternately in the conveying direction **D1**.

[0138] In the present embodiment, when the sheet **Sh1** is conveyed through the conveying path **T1** in the conveying direction **D1**, the sheet **Sh1** is moved relative to the sensor unit **20** including the light irradiation portion **3** and the imaging portion **4**. Therefore, the identification region **R1** to be irradiated with the pattern light **P1** by the light irradiation portion **3** and to be imaged as the identification image **Im1** by the imaging portion **4** moves within the surface **A1** of the sheet **Sh1** along the conveying direction **D1**. Therefore, the conveying direction **D1** is the moving direction of the identification region **R1** within the surface **A1** of the sheet **Sh1**. Therefore, in the present embodiment, at least one of the bright portion **L1** and the dark portion **L2** of the stripe pattern extends along the moving direction of the identification region **R1** within the surface **A1** of the sheet **Sh1**.

[0139] Specifically, the stripe pattern formed by projecting the pattern light **P1** on the surface **A1** of the sheet **Sh1** has a shape in which both the bright portion **L1** and the dark portion **L2** extend along the conveying direction **D1**, which is the moving direction of the identification region **R1**.

However, even if at least one of the bright portion **L1** and the dark portion **L2** is not strictly parallel to the moving direction of the identification region **R1**, the stripe pattern is considered to be along the moving direction of the identification region **R1** as long as the angle between the bright portion **L1** and the dark portion **L2** and the moving direction of the identification region **R1** is within a tolerance (less than 20 degrees). This makes the influence of the asperities on the surface **A1** more likely to appear in the integrated image **Im10**, which is an integration of the identification image **Im1** when the identification region **R1** moves within the surface **A1** of sheet **Sh1**.

[0140] By the way, in the method for obtaining the roughness of the surface **A1** from the shadow image caused by the asperities as in the above-described related technique, for example, in the case of the paper sheet **Sh1**, the local fiber asperities are strongly reflected in the calculation result, so that the calculation result does not necessarily have a linear relationship with the arithmetic average height (**Sa**). Therefore, in the above-described method of the related technique, it is difficult to determine from the calculation result the magnitude of the surface roughness of the sheet **Sh1** of the same type (for example, plain paper), although it may be possible to discriminate between glossy paper with a high flatness (gloss paper) and plain paper, for example. Therefore, in the above-described method of the related technique, in order to determine the magnitude of the surface roughness, it is necessary to prepare in advance, for example, a table (database) in which the calculation results of various sheets **Sh1** are associated with arithmetic average heights (**Sa**).

[0141] In contrast, in the sheet identification device **2** according to the present embodiment, by optimizing the line width, the predetermined angle $\theta 1$, and the like of the pattern light **P1**, the asperity information having a high linearity with the arithmetic average height (**Sa**) can be calculated while also reducing the influence of local fibers. Therefore, according to the method of the present embodiment, it is possible to uniquely obtain an arithmetic average height (**Sa**) from the calculation result of the asperity identification portion **22** without preparing in advance the table (database) in which the calculation results (asperity information) are associated with arithmetic average heights (**Sa**).

[0142] FIG. **14** and FIG. **15** show examples of the identification image **Im1** and the integrated image **Im10** obtained by the sheet identification device **2** according to the present embodiment. Here, as the imaging conditions of the identification image **Im1**, it is assumed that the predetermined angle $\theta 1$ is 40 degrees, the resolution (number of pixels) of the imaging element **41** is 100×100 , the line width **W1** (see FIG. **13**) of the bright portion **L1** of the stripe pattern produced by the pattern light **P1** is $120 \mu\text{m}$, and the line width **W2** (see FIG. **13**) of the dark portion **L2** is $120 \mu\text{m}$. Further, it is assumed that the image magnification **M** is "1", and the extending direction

(longitudinal direction) of each of the bright portion L1 and the dark portion L2 of the stripe pattern produced by the pattern light P1 is the same as the conveying direction D1 (the moving direction of the identification region R1). Here, the identification image Im1 and the integrated image Im10 are shown for a total of four types of sheets Sh1; two types of sheets Sh1 with arithmetic average heights Sa of 0.8 μm and 2.3 μm in FIG. 14, and two types of sheets Sh1 with arithmetic average heights Sa of 5.16 μm and 6.1 μm in FIG. 15. In both FIG. 14 and FIG. 15, the identification image Im1 is shown on the left side, and the integrated image Im10 obtained by integrating the identification image Im1 is shown on the right side.

[0143] The identification image Im1 is composed of a plurality of pixels, and each of the plurality of pixels has a pixel value corresponding to luminance. In the present embodiment, as an example, the relationship between the luminance and the pixel value is defined such that the higher the luminance, the larger the pixel value. Therefore, in the identification image Im1 obtained by capturing the identification region R1 on which the pattern light P1 is projected, the pixel values of the pixels corresponding to the bright portion L1 are relatively large values, and the pixel values of the pixels corresponding to the dark portion L2 are relatively small values.

[0144] The integrated image Im10 is an image obtained by integrating identification image Im1 when the identification region R1 moves by a predetermined number of pixels of the identification image Im1. Specifically, the integrated image Im10 is an image obtained by integrating a predetermined number of pixels of the identification image Im1 in the conveying direction D1, which is the moving direction of the identification region R1, dividing the integrated pixel value of each pixel by the number of pixels for averaging. Here, as an example, the predetermined number of pixels to be integrated in the integrated image Im10 is 50 pixels (pix.). That is, each pixel of the integrated image Im10 has, as a pixel value, an average value of pixel values of 50 pixels (pix.) in the conveying direction D1 in the identification image Im1.

[0145] The upper part of FIG. 14 shows the identification image Im1 and the integrated image Im10 for glossy paper having an arithmetic average height (Sa) of 0.8 μm , that is, having a high flatness, and the lower part of FIG. 14 shows the identification image Im1 and the integrated image Im10 for plain paper having an arithmetic average height (Sa) of 2.3 μm . Similarly, the upper part of FIG. 15 shows the identification image Im1 and the integrated image Im10 for plain paper having an arithmetic average height (Sa) of 5.16 μm , and the lower part of FIG. 15 shows the identification image Im1 and the integrated image Im10 for plain paper having an arithmetic average height (Sa) of 6.1 μm .

[0146] As shown in FIG. 14 and FIG. 15, as the arithmetic average height (Sa) increases, the deformation and distortion of the stripe pattern produced by the pattern light P1 in the identification image Im1 increase. Specifically, when the deformation and distortion of the stripe pattern increase, the boundary line between the bright portion L1 and the dark portion L2 of the stripe pattern is distorted, and the variation in the line width of each of the bright portion L1 and the dark portion L2 increases. That is, as the arithmetic average height (Sa) increases, the undulation component of the height of the surface A1 increases, so that the variation in the line width of each of the bright portion L1 and the dark portion L2 increases. As described above, it is apparent in FIG. 14 and FIG. 15 that, in the identification image Im1, deformation, distortion, or the like occurs in the stripe pattern produced by the pattern light P1 in accordance with the asperities on the surface A1.

[0147] Further, as shown in FIG. 14 and FIG. 15, in the integrated image Im10, as the arithmetic average height (Sa) increases, the line width of an intermediate layer L3 on the boundary line between the bright portion L1 and the dark portion L2 increases. Specifically, as the deformation and distortion of the stripe pattern produced by the pattern light P1 in the identification image Im1 increase, the boundary line between the bright portion L1 and the dark portion L2 of the stripe pattern is distorted, so that the line width of the intermediate layer L3 generated by integrating (averaging) the boundary lines increases (becomes thick). As described above, in FIG. 14 and FIG. 15, it is apparent that the intermediate layer L3 having a line width corresponding to the asperities

on the surface A1 is generated in the integrated image Im10. In the sheet identification device 2 according to the present embodiment, the asperity information of the surface A1 of the sheet Sh1 is identified by analyzing the width (line width) of the intermediate layer L3 of the integrated image Im10.

[0148] In particular, in the present embodiment, an image obtained by integrating the identification image Im1 when the identification region R1 moves by a certain amount of movement is the integrated image Im10, and the amount of movement is equal to or greater than a value obtained by dividing the pixel pitch of the imaging portion 4, which captures the identification image Im1, by the image magnification M. In the present embodiment, since it is assumed that the image magnification M is “1”, the amount of movement of the identification region R1 is equal to or greater than the pixel pitch of the imaging portion 4. Thus, the integrated image Im10 is obtained by integrating the identification image Im1 on the sheet Sh1 which has moved by at least the pixel pitch, due to the movement of the identification region R1 by the amount of movement. Therefore, in the integrated image Im10, the intermediate layer L3 corresponding to the variation in the line width of each of the bright portion L1 and the dark portion L2 is likely to occur.

[0149] However, as shown as the integrated image Im10 in the lower part of FIG. 15, the width (line width) of the intermediate layer L3 may not be uniform throughout the entire integrated image Im10 in the conveying direction D1. This is because the variation in the line width of each of the bright portion L1 and the dark portion L2 is not evened out even in the integrated image Im10, and the undulation component of the line width of each of the bright portion L1 and the dark portion L2 caused by the asperities on the surface A1 remains. For example, as in the case of the paper sheet Sh1, since a undulation component or the like generated by continuous entanglement of a large number of fibers occurs in a relatively long cycle, the line width of each of the bright portion L1 and the dark portion L2 also varies in the integrated image Im10 in accordance with such a long-cycle asperities. Therefore, in order to make the width of the intermediate layer L3 uniform throughout the entire integrated image Im10 in the conveying direction D1, it is preferable that the amount of movement of the identification region R1 for obtaining the integrated image Im10 have a long asperity cycle (distance) or more. Thus, the integrated image Im10 is obtained by integrating the identification image Im1 over a range equal to or longer than the long asperity cycle (distance), so that the width of the intermediate layer L3 can be made uniform throughout the entire integrated image Im10 in the conveying direction D1.

[0150] FIG. 16 shows the pixel value of each pixel for the above four types of sheets Sh1, focusing on any position (one row) of the integrated image Im10 in the conveying direction D1 when the predetermined number of pixels to be integrated is equal to or greater than the long asperity cycle of the sheet Sh1. In FIG. 16, the horizontal axis represents the pixel number (pixel position) in the “row direction” which is the arrangement direction of the bright portion L1 and the dark portion L2 in the integrated image Im10 (the left-right direction in FIG. 14 and FIG. 15), and the vertical axis represents the pixel value (luminance value). That is, according to FIG. 16, there is a region corresponding to the intermediate layer L3 between the bright portion L1 having pixel values of “50” or more and the dark portion L2 having pixel values of less than “30”. The width of the intermediate layer L3 corresponds to the slope of the rise (or fall) of the pixel values between the bright portion L1 and the dark portion L2, and a gentler slope indicates a larger width of the intermediate layer L3. In other words, the larger (rougher) the arithmetic average height Sa of the surface A1 of the sheet Sh1, the gentler the slope of the rise (or fall) of the pixel values between the bright portion L1 and the dark portion L2. Namely, among the graphs of the four types of sheets Sh1 shown in FIG. 16, the slope of the graph whose arithmetic average height Sa is “6.1” is the most gentle.

[0151] Then, the width of the intermediate layer L3 is extracted from the graph of FIG. 16 and a representative value (for example, an average value, a mode value, a median value, or the like) of the widths of a plurality of intermediate layers L3 is calculated for each integrated image Im10,

whereby the width of the intermediate layer L3 in the integrated image Im10 can be calculated. The width of the intermediate layer L3 calculated in this way, that is, the average value of the slope width of the graph in FIG. 16 is as shown in FIG. 17 when expressed in the number of pixels. In FIG. 17, the horizontal axis represents the arithmetic average height Sa, and the vertical axis represents the width (average value of the slope widths) of the intermediate layer L3. In the example of FIG. 17, it can be confirmed that there is a high linearity between the width of the intermediate layer L3 of the integrated image Im10 and the arithmetic average height Sa for at least the above four types of sheets Sh1.

[0152] As described above, by identifying the asperity information using the integrated image Im10 obtained by integrating identification image Im1, the asperity information can be more easily identified even if there is “blurring” in the identification image Im1. That is, in the imaging portion 4, which captures the identification image Im1, a relatively long exposure time is required to capture the identification image Im1 as bright as possible, and the movement of the subject (sheet Sh1) may cause “blurring” in the identification image Im1.

[0153] As an example, when the conveying speed of the sheet Sh1 is set to 500 mm/sec to improve the productivity of the image processing apparatus 10, and the exposure time is 10 ms, which is the maximum, in the imaging portion 4 with a frame rate of 100 Hz, the sheet Sh1 moves 5 mm during the exposure time. Therefore, “blurring” is caused in the identification image Im1 by the movement of the sheet Sh1, and with the method for obtaining the roughness of the surface A1 from the shadow image of the identification image Im1 as in the related technique, it is difficult to obtain the roughness of the surface A1 because the shadow is crushed by the “blurring”. When the related technique described above is applied, in order to suppress the “blurring” of the identification image Im1, it is necessary to decrease the conveying speed of the sheet Sh1 or shorten the exposure time by using a high-sensitivity imaging element 41 for the imaging portion 4. This causes problems such as a decrease in productivity of the image processing apparatus 10 and an increase in cost of the imaging portion 4. On the other hand, in the sheet identification device 2 according to the present embodiment, since the asperity information is identified using the integrated image Im10 which can include a “blurring” component in the first place, it is not necessary to suppress “blurring” of the identification image Im1. Therefore, in the present embodiment, it is easy to improve the productivity of the image processing apparatus 10 and to reduce the cost of the imaging portion 4.

[0154] Further, in the present embodiment, the amount of movement of the identification region R1 for obtaining the integrated image Im10 is equal to or greater than the long asperity cycle of the sheet Sh1. The amount of movement is defined by at least one of the conveying speed of the sheet Sh1 and the exposure time of the imaging portion 4, which captures the identification image Im1. Thus, the width of the intermediate layer L3 is uniform throughout the entire integrated image Im10 in the conveying direction D1, making it possible to obtain the width of the intermediate layer L3 even when focusing on any position (one line) of the integrated image Im10 in the conveying direction D1. As a result, the calculation load for calculating the width of the intermediate layer L3 can be reduced.

[0155] Next, specific processing for identifying the asperity information based on the identification image Im1 by the asperity identification portion 22 will be described with reference to FIG. 18. Steps S1, S2, . . . in the flowchart shown in FIG. 18 represent the numbers of the processing procedure (steps) executed by the control portion 16. The processing to be described below is started in accordance with the timing at which the sheet Sh1 passes the position (monitor position) corresponding to the sensor unit 20 of the conveying path T1, for example.

[0156] When the purpose is to restore a three-dimensional shape, the analysis of the identification image Im1 including the pattern light P1 can be realized by, for example, a method of continuously projecting a plurality of pattern lights P1 and using a Fourier transform or the like of the identification images Im1 to calculate a phase change of the pattern light P1. However, this method

has a relatively high calculation load, takes a relatively long time to calculate the roughness (asperity information) of the surface A1, and also requires relatively high hardware (CPU, GPU, memory, etc.) costs. Therefore, in the present embodiment, instead of the above-described method, the following method is adopted so that the roughness (asperity information) of the surface A1 can be calculated by relatively simple arithmetic processing.

[0157] That is, in the present embodiment, the asperity identification portion 22 calculates the width (line width) of the intermediate layer L3 by focusing on any one row (one line) of the integrated image Im10 with the arrangement direction of the bright portion L1 and the dark portion L2 in the integrated image Im10 (the left-right direction in FIG. 14 and FIG. 15) as the “row direction”. That is, the integrated image Im10 is an “N pixels×M rows” image in which N pixels are arranged in the arrangement direction of the bright portion L1 and the dark portion L2, and the asperity identification portion 22 calculates the line width of any row of the M rows included in the integrated image Im10. The asperity identification portion 22 calculates the line width of the intermediate layer L3 calculated by focusing on one row as the asperity information. Thus, the asperity identification portion 22 can obtain asperity information having a correlation with the arithmetic average height (Sa) of the surface A1, based on the line width of the intermediate layer L3.

Step S1

[0158] Specifically, in step S1, the control portion 16 determines whether the sheet Sh1 has reached the monitor position, that is, the position corresponding to the sensor unit 20 on the conveying path T1. When the sheet feed portion 14 supplies the sheet Sh1 to the image forming portion 13, the control portion 16 determines that the sheet Sh1 has reached the monitor position when the sheet Sh1 is detected by a sensor at the monitor position (S1: Yes), and shifts the processing to step S2. On the other hand, if the sheet Sh1 is not detected by the sensor at the monitor position, the control portion 16 determines that the sheet Sh1 has not reached the monitor position (S1: No), and shifts the processing to step S1.

Steps S2 and S3

[0159] In step S2, the control portion 16 controls, at the acquisition portion 21, the light irradiation portion 3 to cause the light irradiation portion 3 to apply the pattern light P1. Thus, the pattern light P1 is projected on the identification region R1 of the surface A1 of the sheet Sh1. In step S3, the control portion 16 controls, at the acquisition portion 21, the imaging portion 4 to cause the imaging portion 4 to image the identification region R1 on which the pattern light P1 is being projected. Thus, an identification image Im1, which is an image of the identification region R1 of the surface A1 of the sheet Sh1, is generated by the imaging portion 4.

Step S4

[0160] In step S4, the control portion 16 integrates the identification image Im1 when the identification region R1 moves by a certain amount of movement within the surface A1 of the sheet Sh1 to generate the integrated image Im10. Specifically, the time required for the identification region R1 to move by a certain amount of movement (for example, 5 mm) within the surface A1 is set as the exposure time of the imaging portion 4 and then imaging is performed by the imaging portion 4, the image acquired by the acquisition portion 21 is the integrated image Im10 after integration. In the present embodiment, the amount of movement of the identification region R1 for obtaining the integrated image Im10 is equal to or greater than the long asperity cycle of the sheet Sh1. The generation of the integrated image Im10 may be performed in the control portion 16.

Step S5

[0161] In step S5, the control portion 16 acquires, at the acquisition portion 21, an image of one row (one line) of the integrated image Im10 from the imaging portion 4. That is, the acquisition portion 21 acquires the integrated image Im10 of one row corresponding to one pixel in the column direction. Since the imaging portion 4 (imaging element 41) is generally designed to sequentially read out an image for each row, the amount of memory used can be kept low by acquiring and

analyzing (steps S6 and S7) the integrated image Im10 for each row in this manner. The row to be acquired at this time may be predetermined or set by the user.

Step S6

[0162] In step S6, the control portion 16 executes, at the acquisition portion 21, preprocessing on the integrated image Im10. At this time, one row (one line) of the integrated image Im10 acquired in step S5 is subjected to the preprocessing. That is, the control portion 16 executes the preprocessing on the integrated image Im10 row by row. The preprocessing includes, for example, filtering processing. Specifically, the control portion 16 performs noise removal or the like on one row of the integrated image Im10 in the filtering processing. The preprocessing may include trimming processing for cutting out only a part of the integrated image Im10 to narrow down the area to be processed in step S7. In addition, the filtering processing and the like are not essential, and may be omitted as appropriate.

Step S7

[0163] In step S7, the control portion 16 extracts, at the asperity identification portion 22, width data indicating the width (line width) of the intermediate layer L3 from the integrated image Im10. At this time, the width data is extracted from one row (one line) of the integrated image Im10 acquired in step S5. That is, the control portion 16 executes extraction of width data on the integrated image Im10 row by row. Specifically, with respect to one row of the integrated image Im10, the control portion 16 classifies each pixel as the bright portion L1 and the dark portion L2 by regarding a pixel having a pixel value equal to or greater than a first threshold value (for example, “50”) as the bright portion L1 and a pixel having a pixel value less than a second threshold value (for example, “30”) as the dark portion L2. Here, a pixel which does not correspond to either the bright portion L1 or the dark portion L2, that is, a pixel having a pixel value less than the first threshold value and equal to or greater than the second threshold value, is classified as the intermediate layer L3. Then, the control portion 16 calculates the number of pixels classified as the intermediate layer L3 as width data. At this time, the control portion 16 extracts the number of pixels corresponding to the intermediate layer L3 throughout one row of the integrated image Im10, thereby extracting the sum of the line widths of a plurality of intermediate layers L3.

[0164] The first threshold value and the second threshold value used for classifying the pixels as the bright portion L1, the dark portion L2, and the intermediate layer L3 are, for example, a value determined based on an average value of a plurality of pixels, a value determined in advance (predetermined value), or the like. The second threshold value is smaller than the first threshold value. Alternatively, the control portion 16 may extract the line width for each intermediate layer L3 by extracting the number of pixels consecutive in the row direction corresponding to the intermediate layer L3. In this case, the control portion 16 may use, as the width data, not only the total width length of the plurality of intermediate layers L3, but also the line width of each of the plurality of intermediate layers L3 or a representative value (for example, an average value, a mode value, or a median value) of the line widths of the plurality of intermediate layers L3.

Step S8

[0165] In step S8, the control portion 16 calculates, at the asperity identification portion 22, the arithmetic average height Sa from the width data of the intermediate layer L3. That is, there is a high linearity between the width (line width) of the intermediate layer L3 in the integrated image Im10 and the arithmetic average height Sa, so that the arithmetic average height Sa can be uniquely obtained from the width of the intermediate layer L3 by using a linear regression model.

Step S9

[0166] In step S9, the control portion 16 determines, at the condition determination portion 23, image processing conditions. That is, the condition determination portion 23 determines the image processing conditions including the image forming conditions in accordance with the arithmetic average height Sa calculated in step S8. As an example, when the arithmetic average height Sa increases, the condition determination portion 23 sets the image forming conditions so as to

increase the fixing temperature, decrease the conveying speed, or increase the transfer voltage. Thus, when an image is formed on the sheet Sh1 by the image forming portion 13, the image forming conditions corresponding to the asperities on the surface A1 of the sheet Sh1 are automatically applied.

[0167] The procedure of the sheet identification method described above is merely an example, and the order of the processes shown in the flowchart of FIG. 18 may be changed as appropriate.

[0168] Further, it is not essential that at least one of the bright portion L1 and the dark portion L2 of the stripe pattern extends along the moving direction of the identification region R1 within the surface A1 of the sheet Sh1 (conveying direction D1), and for example, both of the bright portion L1 and the dark portion L2 may be orthogonal to the conveying direction D1. Even in this case, as the identification region R1 moves within the surface A1, the pattern light P1 projected on the surface A1 of the sheet Sh1 is dynamically deformed or distorted according to the asperities on the surface A1. Therefore, even when such pattern light P1 is used, the asperity identification portion 22 can identify the asperity information based on the integrated image Im10 in the same manner as in the third embodiment.

[0169] In the third embodiment, as the method for integrating the identification image Im1, a method of temporally integrating the identification image Im1 is adopted, but the present disclosure is not limited to this, and the identification image Im1 may be spatially integrated. That is, “image integration” also includes integration (addition) of pixel values for each pixel of the image (identification image Im1) in the identification region R1 while moving the identification region R1 in one image. That is, by integrating the pixel values of the image (identification image Im1) in the identification region R1 cut out while the identification region R1 moves by a certain amount of movement in one image, the integrated image Im10, which is an integration of the identification image Im1 when the identification region R1 moves by a certain amount of movement, is obtained.

Claims

1. A sheet identification device comprising: an acquisition portion configured to acquire an identification image which is an image of an identification region on which pattern light is projected of a surface of a sheet on which image formation or image reading is performed; and an asperity identification portion configured to identify asperity information on asperities on the surface of the sheet, based on the identification image.
2. The sheet identification device according to claim 1, further comprising: a condition determination portion configured to determine an image processing condition on the image formation or the image reading, based on the asperity information identified by the asperity identification portion.
3. The sheet identification device according to claim 1, wherein a first imaginary straight line connecting a light irradiation portion configured to apply the pattern light and a center of the identification region is inclined at a predetermined angle with respect to a second imaginary straight line extending along a conveying direction of the sheet.
4. The sheet identification device according to claim 3, wherein the predetermined angle is 20 degrees or more and 90 degrees or less.
5. The sheet identification device according to claim 1, wherein a light irradiation portion configured to apply the pattern light includes: a light source; and a shield configured to block part of light output from the light source to allow the pattern light to pass therethrough.
6. The sheet identification device according to claim 1, wherein the pattern light forms a stripe pattern in which a bright portion and a dark portion are alternately arranged on the identification region.
7. The sheet identification device according to claim 6, wherein the stripe pattern includes a first stripe pattern and a second stripe pattern orthogonal to each other, and a width of at least one of the

bright portion and the dark portion is different between the first stripe pattern and the second stripe pattern.

8. The sheet identification device according to claim 6, wherein a width of at least one of the bright portion and the dark portion is 60 μm or more and 500 μm or less.

9. The sheet identification device according to claim 1, wherein the asperity information includes information on at least one of a dimension of the asperities on the surface in a direction orthogonal to a plane along the surface and a dimension of the asperities on the surface in a direction along the plane.

10. The sheet identification device according to claim 1, wherein the asperity identification portion identifies the asperity information, based at least on a variation in a line width of the pattern light on the identification region.

11. An image processing apparatus comprising: the sheet identification device according to claim 1; and an image processing portion configured to execute at least one of the image formation and the image reading on the sheet.

12. A sheet identification method comprising: acquiring an identification image which is an image of an identification region on which pattern light is projected of a surface of a sheet on which image formation or image reading is performed; and identifying asperity information on asperities on the surface of the sheet, based on the identification image.

13. A sheet identification device comprising: an acquisition portion configured to acquire an identification image which is an image of an identification region on which pattern light is projected of a surface of a sheet on which image formation or image reading is performed; and an asperity identification portion configured to identify asperity information on asperities on the surface of the sheet, based on an integrated image obtained by integrating the identification image when the identification region moves by a certain amount of movement within the surface of the sheet.

14. The sheet identification device according to claim 13, further comprising: a condition determination portion configured to determine an image processing condition on the image formation or the image reading, based on the asperity information identified by the asperity identification portion.

15. The sheet identification device according to claim 13, wherein the amount of movement is defined by at least one of a conveying speed of the sheet and an exposure time of an imaging portion configured to capture the identification image.

16. The sheet identification device according to claim 13, wherein the amount of movement is equal to or greater than a value obtained by dividing a pixel pitch of an imaging portion configured to capture the identification image by an image magnification.

17. The sheet identification device according to claim 13, wherein a light irradiation portion configured to apply the pattern light includes: a light source; and a shield configured to block part of light output from the light source to allow the pattern light to pass therethrough.

18. The sheet identification device according to claim 13, wherein the pattern light forms a stripe pattern in which a bright portion and a dark portion are alternately arranged on the identification region.

19. The sheet identification device according to claim 18, wherein at least one of the bright portion and the dark portion extends along a moving direction of the identification region within the surface of the sheet.

20. The sheet identification device according to claim 13, wherein the asperity information includes information on at least one of a dimension of the asperities on the surface in a direction orthogonal to a plane along the surface and a dimension of the asperities on the surface in a direction along the plane.

21. The sheet identification device according to claim 13, wherein the asperity identification

portion identifies the asperity information, based at least on a variation in a line width of the pattern light on the identification region.
